ธนาคารแห่งประเทศไทย
BANK OF THAILAND

MONETARY POLICY REPORT

June

2019



Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
June 2019

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Kanit Sangsubhan	Member
Mr. Subhak Siwaraksa	Member
Mr. Somchai Jitsuchon	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2019. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Executive Summary

Monetary Policy Conduct in the Second Quarter of 2019

In the Committee's view, the Thai economy was expected to expand at a slower pace than the previous assessment as merchandise and services exports would likely grow at a significantly lower rate than previously expected. Headline and core inflation were projected to be in line with the previous forecast. However, there were pockets of risks that might pose vulnerabilities to financial stability in the future. **Thus, the Committee unanimously voted at the meetings on May 8 and June 26, 2019, to maintain the policy rate at 1.75 percent.** The Committee viewed that the current accommodative monetary policy stance contributed to the continuation of economic growth and was appropriate given the inflation target. Moreover, the Committee expressed concerns over the baht appreciation which might not be consistent with economic fundamentals and would continue to closely monitor developments of exchange rates and capital flows. In the period ahead, there remained a need to address financial stability risks through a combination of tools, including the appropriate policy rate as well as microprudential and macroprudential measures which would need to place greater emphasis on debt serviceability of borrowers.

Against the backdrop of heightened uncertainties surrounding both external and domestic factors going forward, the current accommodative monetary policy stance was deemed appropriate. The Committee would continue to closely monitor developments of economic growth, inflation and financial stability, together with associated risks, especially the impacts of trade tensions, in deliberating appropriate monetary policy in the period ahead.

Assessment of the Economic and Financial Outlook as the Basis for Policy Formulation

1. Global Economy

The global economy was expected to grow at slower rate mainly due to escalated trade tensions between the U.S. and China, causing a decline in global trade volume and investor confidence. However, strong labor markets and consumer confidence as well as government stimulus measures in advanced economies would be key supporting factors for economic growth in the period ahead. **The Committee thus revised the growth projection of Thailand's trading partner economies downward to be 3.0 percent in 2019 and 2.8 percent in 2020. Nonetheless, there remained possibilities that trading partner economies could underperform the baseline projection** due to uncertainties surrounding trade protectionist measures that might intensify, geopolitical risks, and increased risks of the no-deal Brexit.

Monetary policy stance of most central banks were more accommodative. The U.S. Federal Reserve (Fed) would likely cut the policy rate once in 2019 with an additional cut in 2020, compared with the previous assessment of one policy rate hike in 2019. The European Central Bank (ECB) would likely hold its policy rate until the first half of 2020 and launch a plan to inject liquidity in the financial system through the third Targeted Longer-term Refinancing Operations (TLTROs III). Meanwhile, the Bank of Japan (BOJ) was expected to keep both short- and long-term interest rate targets until at least the first half of 2020. Regarding central banks with policy rate cuts, some had raised their policy rates multiple times last year, such as the Bangko Sentral ng Pilipinas (BSP) and the Reserve Bank of India (RBI). On the other hand, other central banks still maintained their policy rates such as Bank of Korea (BOK) and Bank Indonesia (BI).

Emerging markets (EMs) experienced volatile capital flows. Capital outflows from EMs occurred amid intensified trade tensions, driven by both increases in import tariffs as retaliatory measures between the U.S. and China and the U.S. announcement to sanction China's major technology company due to security issues. However, capital flows returned to EMs, especially Thailand and India, following a more dovish monetary policy stance of central banks in advanced economies. In addition, in the case of Thailand, other supporting factors included a greater clarity pertaining to domestic political developments and a larger share

distributed to Thai equities in foreign investment funds following an increase in the weights of Thailand's assets in the MSCI and JP Morgan indices.

2. Financial Conditions and Financial Stability

Thailand's financial conditions remained accommodative as reflected by the real policy rate that remained accommodative and at a low level relative to those of other EMs. Thailand's short-term interest rates in the money market and short-term government bond yields were broadly in line with the policy rate. Medium- and long-term government bond yields fell in accordance with the U.S. treasury yields. Private credit moderated, partly due to a partial loan repayment by large corporates after having switched to corporate bond financing. As a result, financing through the bond and equity markets continued to rise. Meanwhile, credit extended to households continued to grow in all loan types. The baht appreciated against the U.S. dollar relative to the end of the previous quarter, especially following intensified trade tensions and a greater clarity pertaining to Thailand's political developments. The real effective exchange rate also appreciated partly due to the baht appreciation against most trading partner currencies.

There were pockets of risks in the financial system that could pose vulnerabilities to financial stability in the future. These included (1) an accumulation of household debt and deterioration of debt serviceability among some group of households and small businesses, (2) the search-for-yield behavior, particularly among saving cooperatives and large corporates, and (3) risks in the property sector including a decline in foreign demand for Thai condominiums in response to the global economic slowdown which could lengthen time to sell existing excess supply. Nevertheless, risks in the property sector were partly addressed through the revised loan-to-value ratio (LTV) measure, as observed in developers' adjustments by delaying launches of new projects and overall housing price index that remained largely unchanged. Moreover, financial institutions would be more cautious in expanding mortgage loans, especially those for second and subsequent residential units acquired by the same purchasers.

3. Economic and Inflation Outlook

The Thai economy was expected to expand at a slower rate of 3.3 and 3.7 percent in 2019 and 2020, respectively, down from the projection in the previous *Monetary Policy Report*. The downward revision was mainly due to external demand, where growth of merchandise and services exports was projected to be lower than the previous assessment following a slowdown of trading partner economies and global trade volume. Moreover, private consumption and investment were expected to moderate from the previous year. Meanwhile, government spending would continue to support growth albeit at a lesser extent than previously assessed.

The value of merchandise exports in 2019 was expected to remain broadly unchanged from the previous year, a downward revision from the previous forecast. This was a result of intensified trade tensions which dampened global trade volume and external demand. However, Thai merchandise exports in the period ahead would be supported by the relocation of production base of some businesses to Thailand and 5G technology-related infrastructure investment plans in many countries, which would promote global trade and investment. This would, in turn, benefit Thailand's exports of electronics parts going forward.

Exports of services were projected to decelerate and exhibit a slower growth than previously assessed due to a decreased number of foreign tourists and lower spending per head. Meanwhile, global economic slowdown would likely result in a moderate outlook of tourists traveling to regional countries including Thailand. However, in the period ahead, extended exemption of visa-on-arrival (VOA) fees until the end of October 2019 and greater tourism promotion by both government and the private sector might partly help offset the impact. Therefore, the projected number of foreign tourists was revised down to 39.9 and 41.3 million in 2019 and 2020, respectively.

Private consumption in 2019 was expected to moderate compared to the high growth last year, but would continue to expand. Supporting factors included sales promotion continuously offered by businesses, as well as government measures which helped support household consumption to some extent. However, private consumption in the period ahead would be restrained by non-farm household income given signs of moderation in earnings, particularly in export-related manufacturing sector, and lower consumer confidence among low-income households due to elevated household debt.

Private investment was expected to continue expanding despite at a lower rate than the previous assessment following a softer outlook of exports and a decline in investment confidence of businesses. However, private investment in 2020 was projected to improve, thanks to a greater clarity of some mega-investment projects, outlook for additional relocation of production base to Thailand especially industries affected by trade protectionist measures, and the progress on infrastructure development in the Eastern Economic Corridor (EEC).

Public spending was projected to support economic growth at a lesser extent than previously assessed. The downward revision was mainly due to public investment. Investment by the central government decreased due to an expected delay in the enactment of the Annual Budget Expenditure Act, B.E. 2563 (A.D. 2020) following a delayed formation of the new government. Moreover, investment by state-owned enterprises also declined from postponement of some investment projects due to contract reviews and problems in accessing construction sites, as well as changes in construction plans. Meanwhile, public consumption was expected to moderate slightly due to decelerating trend of personnel compensation expenditure.

Headline inflation was projected to be in line with the previous assessment. Core inflation and fresh food prices were expected to be lower than previously anticipated due to data outturns from March to May 2019. In addition, demand-pull inflationary pressures would likely decline. However, energy prices were projected to trend up, especially in 2019, which would help offset the downward pressures, keeping headline inflation close to the previous forecast. **Therefore, the Committee projected headline inflation to average 1.0 percent for both 2019 and 2020, and core inflation to average 0.7 and 0.9 percent for 2019 and 2020, respectively.**

The growth projection was subject to downside risks due to (1) a larger-than-expected slowdown in global trade volume and impact on Thai merchandise exports in the case of intensifed trade protectionist measures, (2) a lower-than-expected trading partner economic growth, particularly in the event of no-deal Brexit, as well as due to consequences of Chinese economic and financial stability problems and geopolitical risks, (3) a delay in the new government formation which could impact the continuation of government budget disbursement and policy implementation, and (4) a lower-than-expected growth of Chinese tourist figures if the Chinese government were to promote domestic tourism. However, **there was a possibility that the Thai economy would outperform the baseline projection** owing to (1) higher-than-projected growth of Thailand's trading partner economies, should there be additional government stimulus measures or better-than-expected improvements on trade tensions, and (2) higher-than-expected domestic demand as a result of (2.1) a sooner-than-expected implementation of government infrastructure investment projects, PPP projects and private investment, and (2.2) a larger-than-expected contribution from government stimulus measures if the new government could accelerate disbursements or release additional measures to stimulate private spending. Meanwhile, **risks to the forecasts of headline and core inflation were expected to tilt downward** in line with risks to growth projections and assumptions on global crude oil price.

1. The global economy

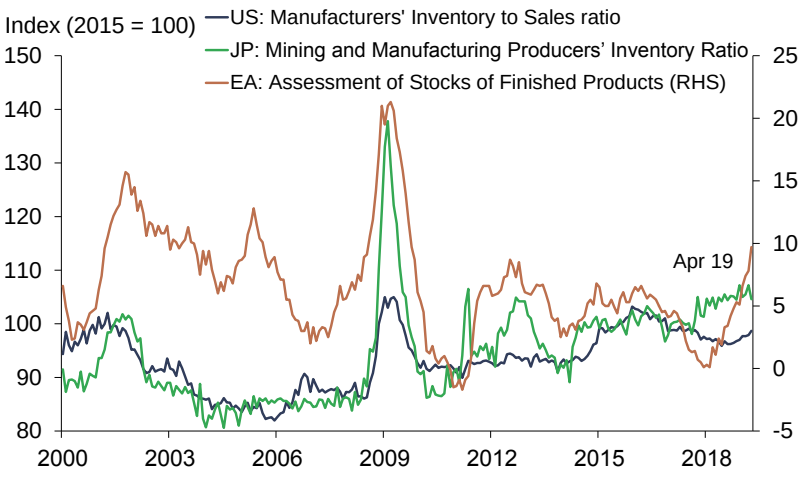
Major advanced economies would slow down as exports and private investment were mainly affected by trade protectionist measures between the U.S. and China. Despite that, economic outturns of many countries in the first quarter of 2019 were higher than expected.

In the first quarter of 2019, most major advanced economies grew higher than expected. The U.S. economy expanded at a faster pace than previously anticipated driven by (1) strong labor market, (2) larger fiscal stimulus, (3) declining imports, as well as (4) inventory accumulation, partly as a result of moderating domestic demand and exports. Similarly, **the Japanese economy** expanded better than expected due to inventory accumulation and imports reduction. Meanwhile, **the euro area** slowed down in line with previous forecast as manufacturing sector and producer confidence continued to weaken, while wages decreased in some countries.

In the period ahead, uncertainties surrounding trade protectionist measures would be the key factor contributing to the continued slowdown of major advanced economies. Merchandise exports would decelerate in tandem with global trade volume. In particular, major auto exporting countries in the euro area might be affected by U.S. tariff measures on auto and auto parts imports. Moreover, private consumption in some countries might be directly affected by trade protectionist measures, especially in the U.S. which could be affected by tariff increases on Chinese imports. Besides, trade uncertainties weighed on investment climate such that investment in major advanced economies were expected to slow down. Meanwhile, the manufacturing sector recovered slowly as business inventory remained high (Chart 1.1). However, some key factors would support economic growth in the period ahead, including (1) robust labor market and high consumer confidence, as well as (2) government stimulus measures in major advanced economies, such as assistance packages for U.S. farmers affected by trade protectionist measures. With regard to private consumption in Japan, there might be an acceleration in spending prior to government's consumption tax increase in October 2019.

Chart 1.1 Inventory in the U.S., Euro Area, and Japan remained high

Inventory to Sales (I/S) ratio (seasonally adjusted)



Note: I/S ratio = Industrial Inventory Index (sa, 2015=100) / Industrial Shipment Index (sa, 2015=100)

Source: Fred, CEIC, European Commission

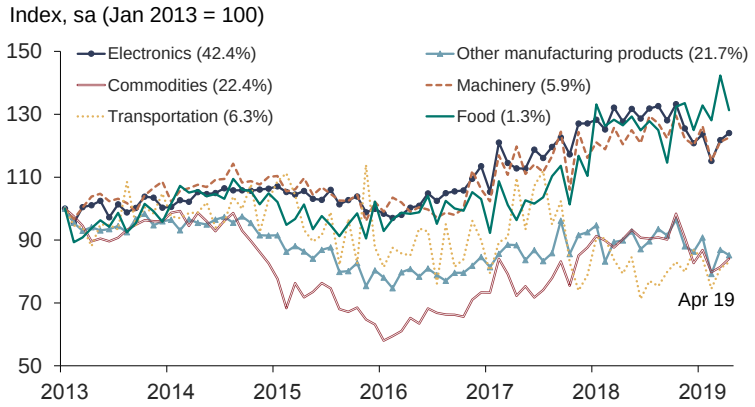
Asian economies would expand at a slower pace mainly due to impacts from trade protectionist measures.

The Chinese economy would grow moderately in 2019, despite improvements in the manufacturing sector and merchandise exports in the first quarter of 2019. Merchandise exports accelerated before the Chinese government reduced the export tax rebate rate from 16% to 13% in April 2019.[1/] This tax rebate reduction was aimed at alleviating impacts on business costs amid economic slowdown from the previous year and uncertainties pertaining to trade protectionist measures in the period ahead. Decelerating global trade volume and investor confidence from impacts of trade protectionist measures would cause a continued slowdown in the manufacturing sector, merchandise exports, and private investment. However, Chinese government's growth-supportive measures, including infrastructure investment and tax cuts, as well as an accommodative monetary policy, would be key factors shoring up the Chinese economy, such that it would not decline significantly and grow according to the government's target.

Asian economies (excluding Japan and China) were expected to moderate. In the first quarter of 2019, many economies experienced a slowdown due to (1) merchandise exports contraction in line with global trade volume (Chart 1.2), as well as (2) country-specific factors including delayed passage of the Philippines' government budget and a decrease in Indonesia's public investment to reduce current account deficit. Looking ahead, Asian merchandise exports would likely slow down in line with global trade volume and the electronics cycle. Meanwhile, softening investor confidence owing to trade protectionist measures might impact manufacturing sector and investment. However, robust labor markets and government stimulus measures, such as wage increases in many countries, would help support domestic demand in the period ahead.

Chart 1.2 Asian exports contracted in line with global trade volume slowdown and country-specific factors

Asian exports value*, classified by product categories



Note: *Asian exports include Hong Kong, Taiwan, South Korea, Malaysia and Singapore. ( ) denotes share of total exports in 2018. Commodity-related products include crude oil, metals, chemicals, rubber, and vegetable oil. Other manufacturing products include textile, papers, furniture, footwear and miscellaneous.

Source: CEIC

[1/] In April 2019, the Chinese government reduced the export tax rebate rate, which was part of the VAT cut measure, from 16% to 13%. The VAT cut measure was aimed at relieving impacts on costs for Chinese businesses amid Chinese economic slowdown and intensifying trade tensions.

Growth projections for Thailand's trading partner economies were revised downward, with risks tilting more to the downside.

Going forward, **Thailand's trading partner economies were projected to slow down more than previously assessed in the previous *Monetary Policy Report*.** This would be mainly due to the decline in global trade volume and investor confidence as a result of the following factors. First, trade protectionist measures intensified given that the U.S. raised tariffs on Chinese imports worth 200 billion U.S. dollars from a previous rate of 10% to 25%, effective on 10 May 2019. Subsequently, China responded by increasing tariffs on U.S. imports worth 60 billion U.S. dollars from 10% to 25%, effective on 1 June 2019. Second, the U.S. Department of Commerce issued regulations prohibiting U.S. companies from dealing businesses with Chinese companies identified as threats to national security in the entity list. This was despite a postponement of regulation enforcement for 90 days. Third, the U.S. might further raise tariffs on the remaining Chinese imports worth at least 300 billion U.S. dollars. In turn, China might retaliate by raising tariffs on the remaining U.S. imports worth 10 billion U.S. dollars. (Box: *The U.S.-China trade war: Implications on the global economy and Thailand*) **The Committee thus revised down growth projection of Thailand's trading partners from 3.1% to 3.0% in 2019 and from 3.0% to 2.8% in 2020.** (Table 1.1)

Table 1.1 Assumption on trading partner growth

Annual change (%YoY)	Weight (%)	2018*	2019	2020
United States	15.9	2.9	2.5 (2.5)	1.7 (1.9)
Euro area	10.3	1.8	1.1 (1.1)	1.4 (1.5)
Japan	14.1	0.8	0.8 (0.9)	0.3 (0.4)
China	17.1	6.6	6.1 (6.2)	5.8 (6.0)
Asia (excluding Japan and China)**	34.2	4.1	3.5 (3.7)	3.5 (3.7)
Total***	100	3.6	3.0 (3.1)	2.8 (3.0)

Note: *Outturn
**Weighted by 7 trading partners' shares in Thailand's exports in 2018, namely, Singapore (5.3%), Hong Kong (7.1%), Malaysia (6.6%), Taiwan (2.2%), Indonesia (5.7%), South Korea (2.8%), and Philippines (4.5%)
***Weighted by 13 trading partners' shares in Thailand's exports in 2018 (including the United Kingdom and Australia)
( ) as reported in Monetary Policy Report, March 2019

The Committee assessed that there were possibilities that trading partners' growth could underperform the baseline projection in line with greater downside risks. Such risks included, first, uncertainties pertaining to trade protectionist measures that could be prolonged and expanded to other countries including Mexico, India, and Europe. Those uncertainties could affect global trade volume and Thailand through supply chain linkages, as well as additionally dampen investor confidence. Despite that, there were some improvements in recent periods after the U.S. announced an indefinite postponement of tariff increases on Mexican imports in return for negotiations about illegal immigration. Second, intensified geopolitical risks might lead to volatilities in financial and commodity markets, including risks from (1) the end of U.S. sanctions exemptions period on Iran's oil exports which

could affect global crude oil supply, (2) unrest in the Strait of Hormuz, as well as (3) denuclearization agreement between the U.S. and North Korea which lacked progress. Third, the risk of no-deal Brexit increased after Boris Johnson, a hard Brexiteer, became the frontrunner in the first round of Conservative party leader election. The elected party leader would replace Theresa May, who resigned as Prime Minister on 7 June 2019.

The global economy was expected to exhibit slower growth due to uncertainties pertaining to trade protectionist measures. Monetary policy of most central banks became more dovish as some central banks cut their policy rates.

Monetary policies of major advanced economies' central banks became more accommodative to prepare and cope with uncertainties regarding potential global economic slowdown and intensified trade protectionist measures. The Federal Reserve (Fed) was expected to cut the policy rate once in 2019 with an additional cut in 2020, instead of gradual policy rate increases as previously anticipated. The European Central Bank (ECB) was expected to keep the policy rate unchanged until the first half of 2020. The possibility of an ECB rate cut also increased in the case of significant economic slowdown. The ECB would also implement a plan to inject liquidity in the financial system through the third targeted longer-term refinancing operations (TLTROs) (beginning September 2019 until March 2021). TLTROs would help businesses to continually access low-cost financing. Meanwhile, the Bank of Japan (BOJ) would still maintain short-term and long-term interest targets until at least the first half of 2020.

Some regional central banks lowered their policy rates after continuous policy rate increases in the previous year. Bangko Sentral ng Pilipinas (BSP) lowered the policy rate given a softer economic outlook and a decrease in inflation towards the $3 \pm 1\%$ target. The Reserve Bank of India (RBI) also lowered the policy rate as the economy was projected to slow down and output gap was expected to be negative. The two central banks cut their policy rates after having continuously increased policy rates in 2018. The policy rates were cut due to concerns over higher-than-target inflation. The cuts were also to safeguard against capital outflow risks following market expectations of a Fed rate hike. In addition, Bank Negara Malaysia (BNM) also cut the policy rate amid signs of economic slowdown from impacts of trade protectionist measures and tightening financial conditions. Reserve Bank of Australia (RBA) cut its policy rate for the first time after keeping the rate unchanged since 2016. This was mainly in response to slow wage rises and lower-than-target inflation. **Other central banks still kept the policy rates unchanged** such as Bank of Korea (BOK) and Bank Indonesia (BI)

Capital outflows from emerging markets (EMs) occurred during periods of intensified trade tensions. However, capital flows returned to EMs after major advanced economies' central banks communicated more dovish monetary policy stance.

In the beginning of the second quarter in 2019, overall risks in the global financial market abated as reflected by the falling VIX index[2/] in April. The decline was driven by several factors (Chart 1.3) including (1) major advanced economies' central banks communicated more dovish monetary policy stance, (2) lower risk of Brexit following the postponement of Brexit process deadline to 31 October 2019, as well as (3) latest U.S. and China's economic outturns being better than market expectations. Investors thus increased portfolio investment in EM assets,

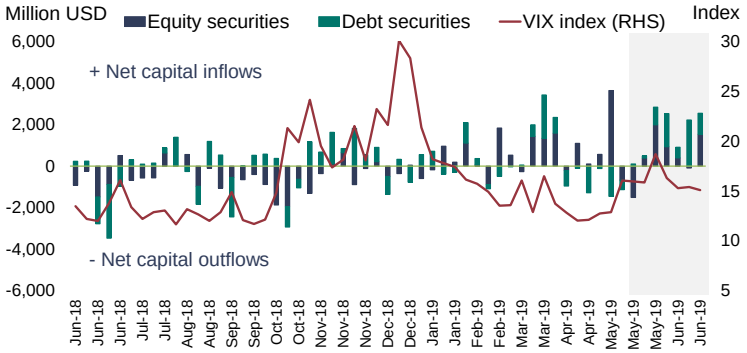
[2/] VIX Index is a measure of the stock market volatility implied by S&P 500 Index options

particularly equities. Later in May, overall risks in the global financial market heightened following intensified trade tensions, from announcements of retaliatory tariff increases and of U.S. sanctions on major Chinese technology companies due to national security concerns. The announcements resulted in the rising VIX index, declines in global stock indices, and capital outflows from EMs. However, in early June, central banks of major advanced economies, especially the Fed and ECB, communicated more dovish monetary policy stance. Capital flows thus returned to EMs, especially Thailand and India. In addition, Thai financial markets were supported by greater clarity pertaining to domestic political situation. Moreover, foreign funds allocated higher investment shares to Thai equities and bonds in tandem with increases in Thailand's weights in the MSCI and JP Morgan benchmark indices.

Looking ahead, the global financial markets would remain highly volatile. EMs capital inflows and outflows may fluctuate in tandem with uncertainties of the ongoing trade protectionist measures, advanced economies' monetary policy stance, Brexit developments, as well as geopolitical risks that might impact commodity prices.

Chart 1.3 EMs experienced net capital outflows during periods of intensified trade tensions. However, capital inflows returned to EMs after major advanced economies' central banks communicated more dovish monetary policy stance.

Net capital inflows to EMs* and VIX index (weekly)



Note: *EMs include Thailand, Indonesia, India, South Africa, and Turkey
Sources: Bloomberg and Institutional Institute of Finance

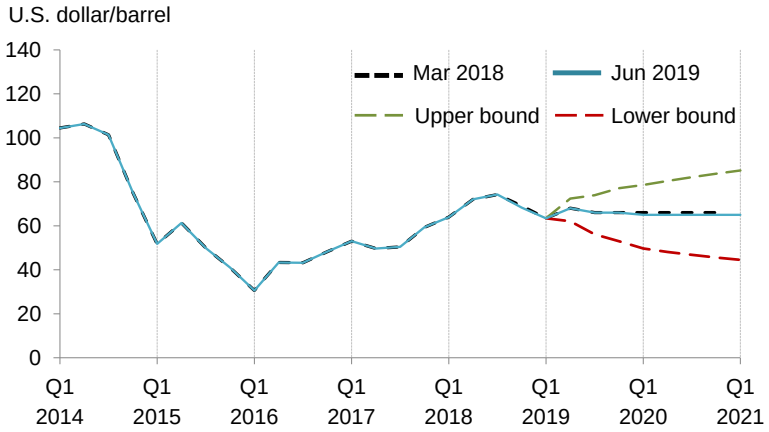
Dubai crude oil prices in the second quarter of 2019 rose from the previous quarter due to geopolitical issues which tightened global crude oil supply. In the period ahead, crude oil prices were expected to fall slightly from lower demand pressures in line with softening global economic outlook, despite tightening crude oil supply.

Dubai crude oil prices in the second quarter of 2019 increased from the previous quarter due to tightened global crude oil supply. The tighter supply was a result of reduced production capacity of OPEC members and allies in accordance with the agreement to cut crude oil production. Other supportive factors included intensified geopolitical risks from U.S. sanctions against Venezuela, the end of U.S. sanctions exemption period for Iran's oil exports, and unrest in the Strait of Hormuz. However, crude oil prices declined since the end of the second quarter of 2019 due to investor concerns about global economic outlook that might be affected by intensifying trade tensions.

The Committee maintained the projection for Dubai crude oil price in 2019 at 65.9 dollars per barrel (Chart 1.4), although crude oil supply tightening was expected as geopolitical

issues in many countries might lead to higher prices. However, crude oil demand was projected to slow down in line with the global economy, and would put downward pressure on prices in the period ahead. **The Committee thus revised down the projection for Dubai crude oil prices in 2020** from 66.0 to 65.0 dollars per barrel. The Committee also assessed that **there were higher possibilities of lower-than-baseline Dubai crude oil prices** compared to the previous *Monetary Policy Report.* This was because the global economy might slow down more than expected from (1) impacts of trade protectionist measures, (2) accelerated U.S. shale-oil production, as well as (3) impacts from the International Maritime Organization (IMO)'s enforcement of the new marine fuel regulations which might lower demand for high-sulfur crude oil. However, crude oil prices might rise given potentially intensifying geopolitical factors, posing impacts on expectations of crude oil production capacities and on investor confidence more than previously assessed.

Chart 1.4 Dubai crude oil prices in 2019Q2 increased from the previous quarter on account of geopolitical factors that led to tighter crude oil supply. Looking ahead, oil prices would slightly decrease due to reduced demand-pull pressures in line with global economic slowdown, although crude oil supply would tighten.



The U.S.-China trade war: Implications on the global economy and Thailand

In recent periods, the U.S. implemented international trade policies more inclined towards protecting its own trading interests, in order to restore its manufacturing sector and support domestic employment. Various measures relating to this included (1) tariff measures on goods imported from many countries, such as tariffs increase on steel and aluminum imports (2) tariff measures on goods imported from certain countries, such as tariffs on 3,607 items imported from China, with a total value of 250 billion U.S. dollars, and (3) non-tariff barriers such as sanctions on major Chinese technology companies for national security reasons. **Such measures weighed on global trade volume and merchandise exports of several countries including Thailand. Moreover, manufacturing, employment, and private investment in related industries were also affected.** This article therefore aims to analyze the impacts of trade protectionist measures on the Thai economy through the following channels.

1. Impacts on merchandise exports

Over recent periods, growth of Thai merchandise exports value continued to decline. After a high expansion of 12.4% in the first half of 2018, exports value in the first quarter of 2019 contracted by 4.0% (Chart 1), which was the first contraction in 11 quarters. This was due to the impacts of trade protectionist measures between the U.S. and China. Although the U.S. did not directly raise tariffs on Thai imports, Thai merchandise exports were nevertheless affected through three main channels as follows.

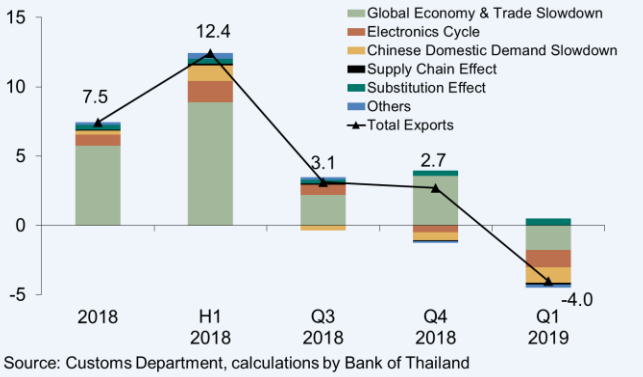
1.1 World trade volume effect

Trade protectionist measures were the main reason for slowdown in global trade volume and Thailand's trading partner economies. This factor contributed 2.9% to the contraction in Thai exports value in the first quarter of 2019 (light green and yellow bars in Chart 1)[3/]. Meanwhile, the International Monetary Fund (IMF) assessed that trade tensions would weigh on confidence and global growth. The IMF thus revised down global trade volume forecast for 2019[4/] from 4.1% to 3.3%. However, developments in the use of trade protectionist measures remained highly uncertain, as this situation

Chart 1 Thailand's exports value growth has decelerated recently.

Contribution to Thailand's exports value growth

%YoY



Source: Customs Department, calculations by Bank of Thailand

[3/] Contribution to growth in Thai exports value can be decomposed into the following channels.

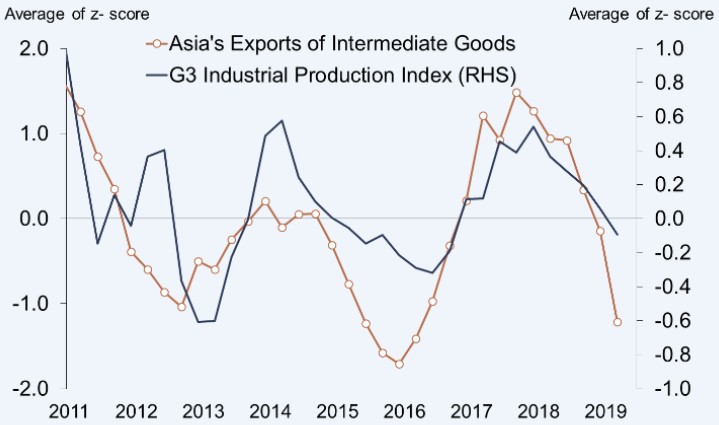
1. Electronics cycle: assessed from Thailand's total electronics exports value (excluding exports to China)
2. Substitution effect: assessed from export values on items for which the U.S. imposed tariffs on China, allowing Thailand to increase market share (compared to 4-year average) at the expense of China
3. Supply chain effect: assessed from Thai exports value that contribute value-added to tariffed Chinese goods exported to the U.S. Value-added were calculated using the structure of inter-country input-output table (ICIO)
4. Chinese demand slowdown: assessed from the residual of Thai exports value to China after accounting for the supply chain effect
5. Slowdown in economic and global trade volume: assessed from exports value of other items not included in the aforementioned categories.

[4/] Forecast as of April 2019

could be prolonged or intensified. As a result, firms delayed production and investment in order to wait for more clarity. This was reflected by continuous declines in the Manufacturing Purchasing Managers' Index (PMI), particularly new orders for capital goods. Deceleration in advanced economies' manufacturing sectors were followed by deceleration in the exports value of Asian economies' intermediate goods (Chart 2). **Moreover, trade protectionist measures partly contributed to a prolonged downturn in the electronics cycle**, as well as sluggish recovery in global and Thai electronics exports. This factor contributed 1.3% to the decline in the Thai exports value in the first quarter of 2019 (orange bar in Chart 1).

Chart 2 Continued deceleration in manufacturing sectors were followed by a decline in Asia's exports of intermediate goods

G3 industrial production and Asia's exports of intermediate goods



Source: Bloomberg, calculations by Bank of Thailand

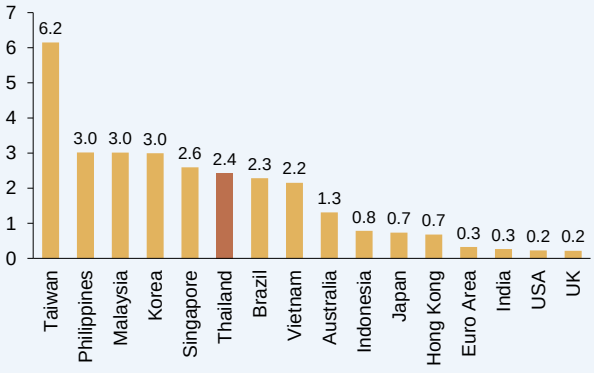
1.2 Supply chain effect

Trade protectionist measures subsequently impinged on the exports of countries integrated in U.S. and China's supply chains. This was because most goods being tariffed were manufacturing goods with high linkages to many countries. Examples of such goods were electronics parts, auto parts, machinery, petrochemical, and chemical products. The production of these exports products required intermediate and capital goods imports from other countries. Countries most affected by this channel were small Asian economies with high linkages to China, namely Taiwan, the Philippines, Malaysia, Korea, Singapore, and Thailand (Chart 3). The supply chain effect contributed 0.1% to the contraction in Thai exports value in the first quarter of 2019 (black bar in Chart 1), with the most affected categories being electronics and optical instruments.[5/]

Chart 3 Asian countries with high linkages to China's supply chains are likely to be highly affected

Sources of value added in China's total exports

% of GDP for each country



Source: OECD TiVA, calculations by Bank of Thailand

[5/] Thailand is confronted with the supply chain effect in both direct and indirect ways. The direct effect is via a decrease in Thai exports to China. The indirect effect is via a fall in Thai exports to other countries, owing to the decline in their exports to China.

1.3 Substitution effect of Thai exports in place of Chinese goods in the U.S. market

A decline in U.S.-China trade may create an opportunity for Thailand to export more to the U.S. market in place of China. Products which benefitted from this channel were auto parts, integrated circuits, and machinery for rubber and plastics production. This factor helped offset a contraction in Thai exports values in the first quarter of 2019 by 0.5% (dark green bar in Chart 1).

Going forward, trade protectionist measures will likely be prolonged or escalated owing to (1) additional tariff measures which the U.S. could levy on the remaining Chinese imports, with a total value of 300 billion U.S. dollars, as well as tariffs on auto and auto parts that the U.S. might impose on other countries, (2) tariffs could be increasingly used as a bargaining tool for non-trade issues, such as illegal immigration between the U.S. and Mexico, and (3) non-tariff barriers (NTBs) could be employed more often, such as the case of the U.S. identifying Chinese technology companies as a threat to national security. Consequently, international trade outlook will remain highly uncertain. It may also be an important risk which might impinge on global trade volume and trading partner economies more than previous assessment by other organizations. Thus, firms should stand ready to make timely adjustments in response to possible changes. For instance, adding product value could help firms to reap greater benefits from substitution effect. Moreover, firms should explore new export destinations to diversify impacts from China's supply chain effect. This could be done by exporting raw materials to other countries which manage to gain shares in U.S. markets in place of China.

2. Impacts on domestic traders

Trade protectionist measures also affected traders in Thailand, as **China may dump products which were not exported to the U.S. in the Thai market. An analysis of imports from China and the similarity index[6/], indicating product categories most prone to Chinese dumping, reveals some signs of dumping from China. However, the signs are still vague.** Chinese imports value accelerated in some product categories, including aluminum structures, air conditioners, vehicle parts, textiles, and electronics. Nonetheless, Chinese imports of these products accounted for less than 1% of total imports value. Nevertheless, should the U.S. decide to impose additional tariffs on Chinese products with a total value of 300 billion U.S. dollars, China may further dump their products. Then, impacts on Thai firms would be more prominent owing to these firms' lack of price competitiveness in these product categories. Meanwhile, domestic consumers would be able to purchase imports at lower prices, but should also consider product quality issues all together.

3. Impacts on domestic manufacturing and employment

In case global trade deteriorated with significant effects on Thai exports, it may consequently lead to production and employment cuts in Thailand's manufacturing sector, particularly the export-related industries. Firms may eventually cut their employment or compensation of employees. In the first five months of 2019, **there were some signs of production and employment cuts, particularly over-time employment in some export-related industries.** However, some of these workers shifted towards services and trade sectors, which partially shored up overall purchasing power and consumption.

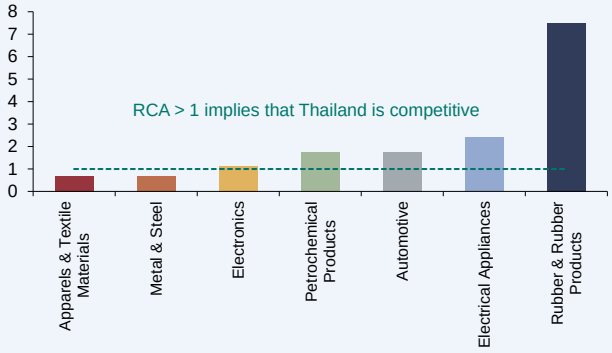
[6/] Similarity index is calculated from $\min\left(\frac{\text{Thai import value of product } a \text{ from China}}{\text{Thai import value of all products from the world}}, \frac{\text{U.S. import value of product } a \text{ from China}}{\text{U.S. import value of all products from the world}}\right)$ x 100, with high index value indicating that product "*a*" stands a high chance of being dumped by China.

4. Impacts on private investment

Trade protectionist measures could benefit some countries thanks to **investment diversion.** Firms may decide to relocate production base away from China to countries with high potential for investment such as Vietnam and Thailand, to avoid impacts of trade tensions. Thai industries including rubber and rubber products, electrical appliances, automotive (including tyres and inner tubes), petrochemical products, and electronics, still maintain competitiveness in the global markets, as reflected by Thailand's revealed comparative advantage (RCA)[7/] indicator (Chart 4). It is therefore possible that more firms in such industries will relocate production base to Thailand in the future. In recent periods, foreign direct investment in the electronics sector accelerated significantly (Chart 5). FDI from China increased continually since the fourth quarter of 2018, particularly rubber products, metals, and chemicals sectors, despite accounting for a small proportion of Thailand's total FDI.

Chart 4 A number of Thai industries such as rubber, rubber products and electrical appliances still maintain their global competitiveness

Revealed Comparative Advantage (RCA), classified by industry

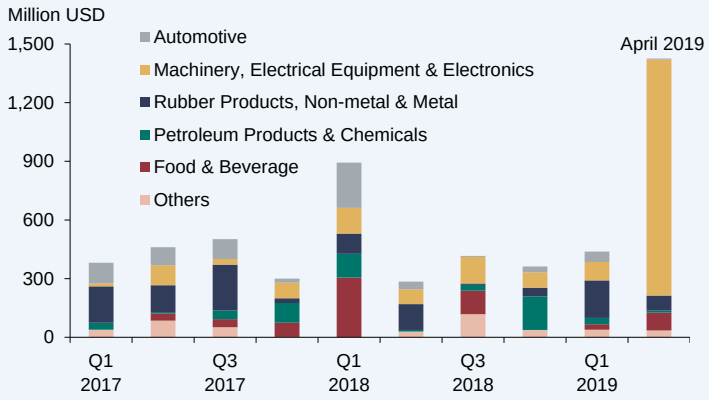


Source: Customs Department, calculations by Bank of Thailand

In summary, trade protectionist measures will likely pose impacts on the Thai economy through various channels, particularly the trade channel via reduction in merchandise exports and weakening confidence in many countries. This partly clarifies the zero growth of Thai exports forecast for 2019 from the high outturns in the previous year. Moreover, impacts through other channels also warrant close monitoring, including (1) Chinese dumping that could negatively affect domestic traders, despite limited impacts in recent periods, (2) production and employment in manufacturing sector, which may subsequently affect household purchasing power, and (3) benefits from foreign firms' relocation of production base to avoid impacts of trade protectionist measures, as some Thai industries maintain competitiveness to attract this FDI.

Chart 5 Over recent periods, there has been some inflows of foreign direct investment in the electronics sector

Gross foreign direct investment inflows, classified by business sector



Source: Bank of Thailand

[7/] RCA is calculated from the proportion of each product's exports value in relation to Thailand's total exports value, divided by the share of such product's exports value in relation to total global exports value.

2. The Thai Economy

2.1 Recent Developments

The Thai economy expanded at a slower pace in the first quarter of 2019, mainly due to external demand which was affected by the slowdown in trading partner economies. Domestic demand continued to support the economic growth.

In **the first quarter of 2019,** the Thai economy expanded 2.8 percent from the same period last year. Growth moderated from the previous quarter and was lower than previously assessed in the previous *Monetary Policy Report,* mainly due to merchandise and services exports. Merchandise exports were affected by the slowdown in trading partner economies, trade protectionist measures between the U.S. and China, as well as the down cycle of electronic products. As a result, exports of manufacturing products, such as electronic components and devices, and electrical appliances, contracted. Exports of other products also declined, including fishery products which experienced raw material shortages, and agricultural products, especially rice as export prices in U.S. dollar term were higher than those of competitors. Meanwhile, services exports contracted in line with a decline in tourism receipts, mostly on account of a decreased number of Chinese and European tourists. However, domestic demand continued to support the economic growth, where private consumption continued to expand despite some moderation in spending growth of semi-durable goods and services. Purchasing power was supported by improved household income and employment, robust consumer confidence, as well as additional government measures aiming to relieve costs of living of the lower income groups through the second phase of the social welfare card scheme. Private investment slowed down in line with investment in residential construction, as well as in machinery and equipment used in offices and manufacturing. Meanwhile, public spending growth accelerated on account of personnel compensation expenditure, while public investment contracted slightly in tandem with lower investment of state-owned enterprises, as investment of state-owned enterprises in this quarter was mostly ongoing investment of existing projects. Overall, the Thai economy expanded 1.0 percent in the first quarter of 2019 relative to the previous quarter, after seasonal adjustment.

The Thai economy continued to gain traction from the previous quarter in **the second quarter of 2019**, as reflected by recent economic indicators. The key growth driver was domestic demand. Private consumption continued to expand across all product categories, thanks to robust purchasing power and additional government measures aiming to relieve costs of living for the lower income groups that were expected to continue supporting household purchasing power. Private investment expanded in line with continued investment in public-private partnership (PPP) projects, including investment diversion of some industries to Thailand to mitigate the impact of trade protectionist measures. Public spending expanded on the back of personnel compensation expenditure and investment that resumed growth this quarter due to accelerated budget disbursements for central government investment following project reviews to comply with the master plans under the National Strategy. Meanwhile, the contraction of merchandise exports moderated for some product categories, such as electronics, thanks to the trade diversion to mitigate the impact of trade protectionist measures, leading to rising benefits to Thailand's merchandise exports. Moreover, some industries that relocated their production base to Thailand in an earlier period started to export some products, including industries related

to vehicle and electronic parts. Exports of agricultural products improved on account of fruit exports. Exports of services contracted mainly on account of a decline in foreign tourist figures, particularly Chinese tourists, partly due to a slowdown of the Chinese economy. Meanwhile, the slowing outlook of the global economy resulted in a more cautious spending of foreign tourists, leading to a lower spending per trip. Nevertheless, government measures to promote tourism, especially the extended exemption of the visa-on-arrival (VOA) fees from April 30, 2019 to the end of October 2019, would help boost foreign tourist figures going forward.

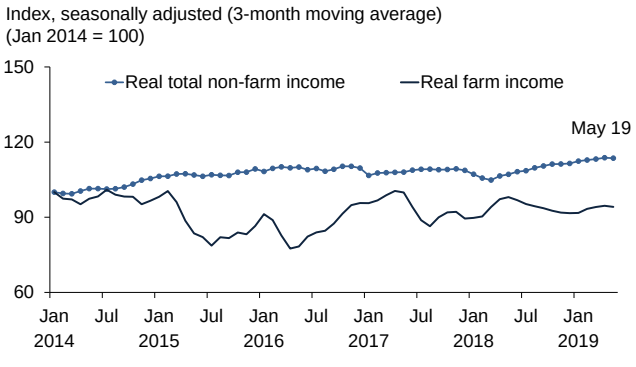
Overall purchasing power continued to improve, mainly on account of employment and income of non-farm households, with signs of a reduction in employment observed in just a few industries. However, purchasing power of farm households warranted close monitoring given intensifying climate variability.

The continued economic expansion in the recent period contributed to sound household purchasing power, both from improvement in income and employment. (Charts 2.1 and 2.2). Income of **non-farm households** continued to improve. Overall employment was stable, supported by employment in the services sector. On the contrary, employment in export-related manufacturing sector, such as auto as well as rubber and plastic industries, started to experience the impact of the export slowdown and a reduction in over-time employment, as well as an adoption of automation.

Employment of **farm households** continued to decline as a result of the drought and migration of some labor to the non-farm sector. Farm income contracted slightly this quarter maily due to a decline in prices, especially prices of sugar cane which declined in line with global sugar prices, and palm oil given a large output to the market. Agricultural output also contracted in line with a lower output of sugar cane than the previous year. Going forward, there remained a need to monitor the strength of farm income. In particular, climate variability must be monitored, as it would not only pose adverse impacts on agricultural sectors globally, Thailand's agricultural sector would also experience a significant impact given its heavy reliance on the climate. (Box: *The impact of climate variability on Thailand's agricultural sector*)

Chart 2.1 Overall purchasing power continued to improve

Household income indicators

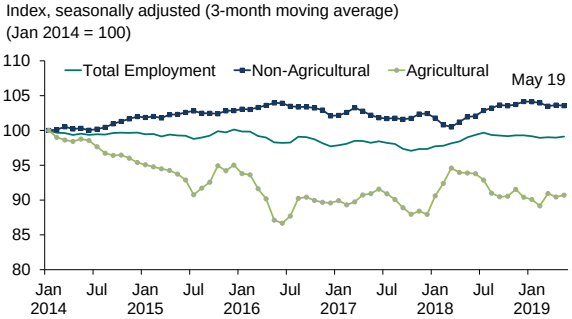


Note: *wage and salary transfer transactions are calculated from 2 databases: Transaction Management a (1) Commercial banks reporting transactions to Bank of Thailand database which covers 90% of all retail transfer transactions and (2) Interbank nd Exchange (ITMX) database which covers 10% of all retail transfer transactions.

Sources: Bank of Thailand, Office of Agricultural Economics, National Statistical Office, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.2 Overall employment remained stable, but employment of farm households continued to contract due to the drought

Employment indicators

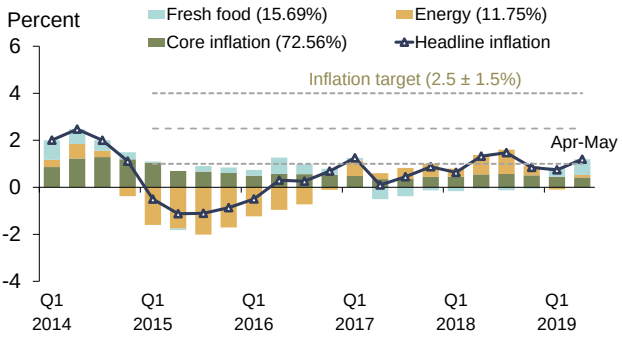


Source: National Statistical Office

Headline inflation increased, particularly fresh food prices. Core inflation decreased slightly in line with core inflation in the food category.

Chart 2.3 Headline inflation increased from the previous quarter mainly due to fresh food and energy prices

Headline inflation and inflation target



Note: ( ) denotes share in inflation baskets
Source: Ministry of Commerce, calculations by Bank of Thailand

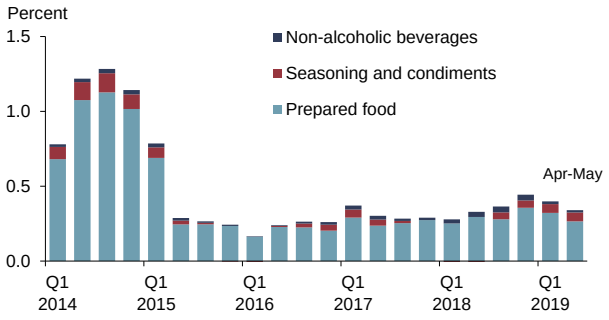
Headline inflation averaged at 1.19 percent over the first two months of the second quarter 2019, an improvement from 0.74 percent in the previous quarter (Chart 2.3). The increase was due to continued increases in fresh food prices in line with vegetable prices, as output was damaged by an abnormally hot weather. The increase was also driven by higher energy prices, particularly as domestic retail oil prices rose in tandem with global crude oil prices.

Core inflation averaged at 0.58 percent over the first two months of the second quarter in 2019, a slight decrease from 0.62 percent in the previous quarter. This was mainly owing to core inflation in the food category, which declined (Chart 2.4) due to slower increases in prices of processed foods. This reflected that an increase in fresh food prices did not sufficiently pass through to prices of processed foods. Meanwhile, core inflation in the non-food categories increased slightly (Chart 2.5), mainly due to increased prices of public transportation services. In other categories, prices slowly increased despite a continued expansion of demand, as structural factors encompassing rising e-commerce trends, intensified price competition and lower production costs from higher production efficiency weighed down core inflation in the non-food categories in the recent periods (Chart 2.1).

Short-term (one- year ahead) inflation expectations according to the survey of businesses in May 2019 stood at 1.9 percent, close to the previous quarter. Inflation expectations of professional forecasters in March 2019 stood at 1.5 percent, a slight increase from 1.3 percent in the previous quarter. Meanwhile, **long-term (five-year ahead) inflation expectations** according to the survey of professional forecasters in April 2019 stood at 1.8 percent, unchanged from the previous survey (October 2018).

Chart 2.4 Core inflation in the food category (28% of core inflation) declined mainly on account of prepared food prices

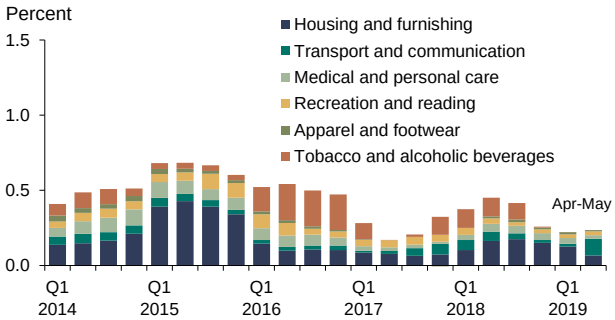
Contribution* to core inflation in the food category



Note: *Contributions to core inflation decompose core inflation into an inflation rate of each component within the core inflation basket, weighted by its corresponding share in the basket.
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.5 Core inflation in the non-food category (72% of core inflation) increased, as a result of rises in public transportation fares

Contribution* to core inflation in the non-food category



Note: *Contributions to core inflation decompose core inflation into an inflation rate of each component within the core inflation basket, weighted by its corresponding share in the basket.
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Table 2.1 Inflation

Annual percentage change	2017				2018				2019	
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Apr - May
Headline Consumer Price Index (Headline CPI)	1.25	0.10	0.45	0.88	0.64	1.31	1.47	0.84	0.74	1.19
Core Consumer Price Index (Core CPI)	0.66	0.47	0.49	0.61	0.61	0.76	0.78	0.71	0.62	0.57
Raw food	0.61	-2.99	-2.25	-0.80	-1.04	-0.35	-0.82	-0.35	2.50	4.29
Energy	6.69	2.67	4.86	5.24	3.01	7.30	9.11	3.39	-0.79	0.92

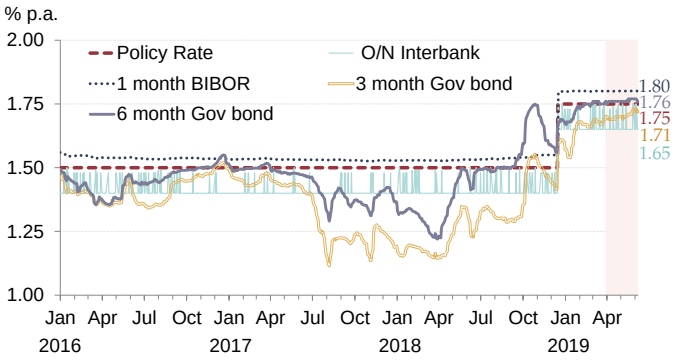
Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Short-term money market rates and short-term government bond yields were stable around the policy rate. Medium-term and long- term government bond yields decreased mainly in tandem with U.S. government bond yields.

In the second quarter of 2019, **short-term money market rates and short-term government bond yields** were stable around the policy rate (Chart 2.6). **Medium- and long-term government bond yields** (Chart 2.7) moved mainly in tandem with U.S. government bond yields. Yields edged up slightly in April, partly owing to improved investor sentiments following better-than-expected U.S. and Chinese economic data outturns. However, from the end of May to June, yields dropped following intensified trade tensions and lower-than-expected U.S. economic outturns, as well as a more dovish stance of monetary policy communicated by the Fed. Moreover, demand, from both domestic and foreign investors, for long-term government bonds which were used as benchmark bonds continued to increase. This partly reflected expectations that overall Thai government bond yields would likely decline. In addition, this was a result of an increase in the weights of Thai bonds in JP Morgan's benchmark index.

Chart 2.6 Short-term money market rates and short-term government bond yields stabilized around the policy rate

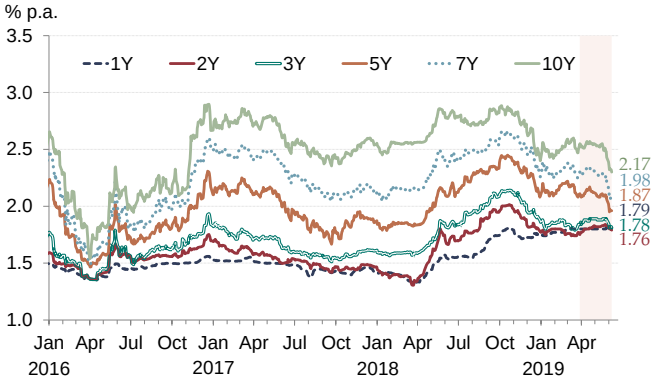
Short-term rates in financial markets



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA) (data as of 25 June 2019)

Chart 2.7 Yields on medium-term and long-term government bonds fell mainly in tandem with U.S. government bond yields, especially towards the end of the quarter.

Government bond yields

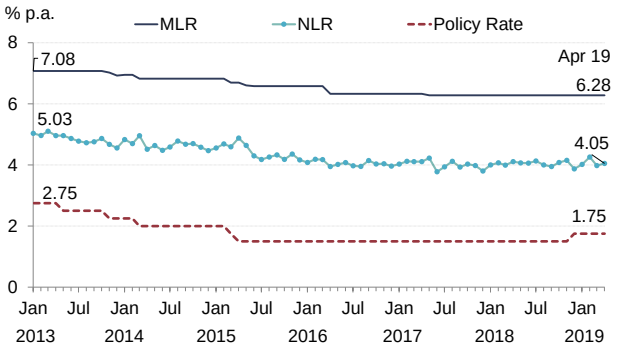


Source: Thai Bond Market Association (Thai BMA) (data as of 25 June 2019)

Overall corporate bond yields declined in the second quarter of 2019 in tandem with government bond yields and credit spread.[8/] **Cost of financing through commercial banks** remained at a low level, as represented in the new loan rate (NLR)[9/] in April 2019 that remained stable at 4.1 percent from the end of the previous quarter (Chart 2.8). Meanwhile, reference loan rates of commercial banks remained mostly unchanged from the previous quarter, with the exception of a slight increase in one medium-sized bank's minimum retail rate (MRR) in May.

Chart 2.8 New Loan Rate (NLR) stabilized at low level

New Loan Rate



Source: Bank of Thailand (data as of April 2019)

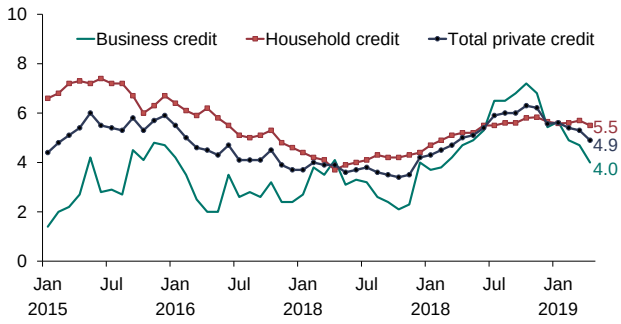
Accommodative financial conditions allowed business financing to continue.

Private credit[10/] moderated in the first quarter and April 2019. In April, private credit expanded 4.9 percent (Chart 2.9) due to a slowdown of business credit, especially in the manufacturing sector. This was partly owing to a partial loan repayment by large corporates from funds obtained through corporate bond financing. On the other hand, credit extended to households continued to grow in all loan types in line with robust expansion of private consumption. In particular, mortgage loan extension accelerated prior to the implementation of the revised loan-to-value (LTV) ratio measures in April.

Chart 2.9 Private credit moderated due to a slowdown in business credit, while household credit continued to expand

Growth of private credit

Percentage change from the same period last year



Note:
1) Private credit includes credit to other depository corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds
2) The data of ODCs credit outstanding to business and household sectors since January 2015 are revised following the improvement of processing system for more accurate and comprehensive ODCs credits data

Source: Bank of Thailand

The net issuance of corporate bonds continued to increase from the previous quarter in the first quarter of 2019, driven by corporate financing of several businesses including businesses in food and beverages, and the IT and telecommunication sectors. This was reflected in net

8/ A credit spread is the difference between corporate and government bond yields with the same tenure, reflecting an assessment on corporate bond issuers' default risks.

9/ NLR is the weighted average of interest rates on new loan contracts extended by 14 Thai commercial banks each month. The loan contracts exclude consumer loans, credit card loans, repurchase agreements, bank guarantees, as well as loans extended to financial intermediaries, the public sector and non-residents. The dataset covers loans with value of 20 million baht or higher for all loan types, purposes and maturities, and includes both secured and unsecured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

10/ Outstanding credit of other depository corporations (ODCs), namely commercial banks, specialized financial institutions, finance companies, savings cooperatives, and market mutual funds.

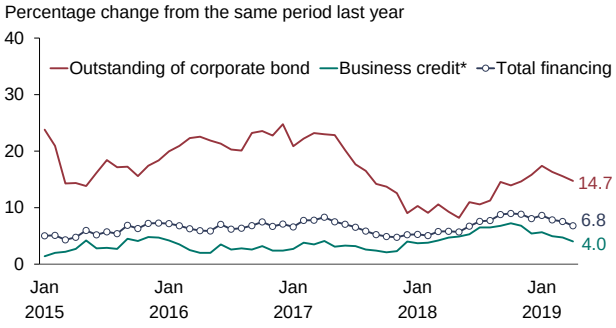
corporate bond outstanding that rose 14.7 percent (Chart 2.10). **Funding through the equity market** continued to increase for businesses related to transport and logistics, and steel production, as well as funding for merger and acquisition in the insurance sector.

Going forward, **financial conditions were expected to remain accommodative.** This was reflected in the real policy interest rate that remained accommodative and moderate compared to other countries (Chart 2.11), despite increasing somewhat due to lower inflation expectations. Meanwhile, costs of business financing through commercial banks, as represented by the new loan rate (NLR), remained stable at a low level. However, financial institutions were expected to tighten credit standards in the second quarter of 2019[11/], particularly for loans extended to large corporates and mortgage loans due to concerns over borrowers' credit quality and outlook of the real estate market.

The baht appreciated against the U.S. dollar and the nominal effective exchange rate (NEER) index also appreciated.

Chart 2.10 Overall corporate financing continued to expand

Growth of corporate bond outstanding and business credit

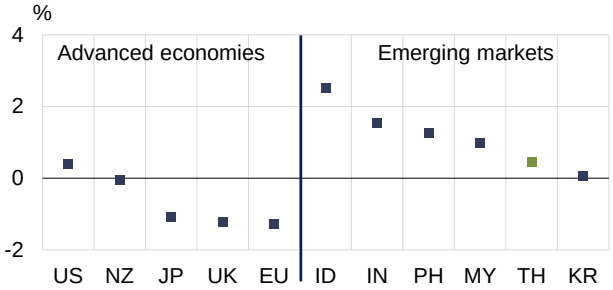


Note: *Business credit covers lending activities of Other Depository Corporation (ODCs)

Sources: Thai Bond Market Association (Thai BMA) and Bank of Thailand

Chart 2.11 Thailand's real policy rate* increased but remained accommodative overall. The rate was relatively low compared with other countries.

Real policy rates*



Note: *Calculated from policy rate subtracted by one-year-ahead inflation expectation according to a survey by Consensus Economics (data as of 10 June 2019)

Sources: Bloomberg, Consensus Economics, calculation by Bank of Thailand

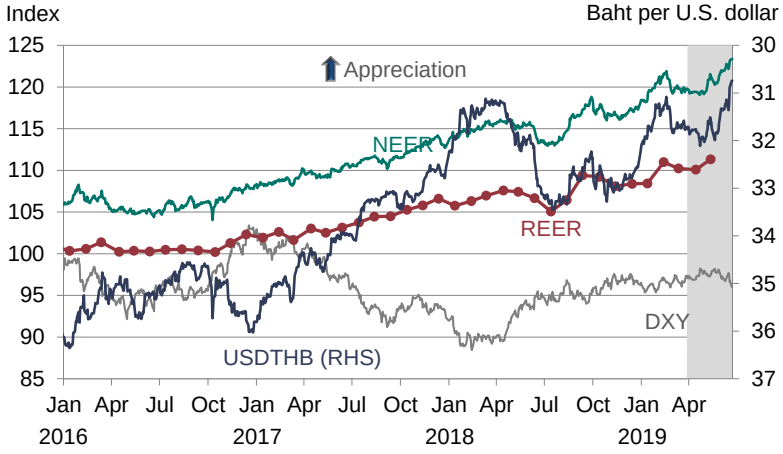
In the second quarter of 2019, **the baht appreciated against the U.S. dollar** relative to the end of the previous quarter (Chart 2.12). In April, the baht depreciated against the U.S. dollar which strengthened following better-than-expected U.S. economic data outturns and increased market concerns over the European economic outlook as economic data outturns continued to turn out lower than market expected. In addition, foreign investors reduced their holdings of EM assets, particularly assets of net oil importing countries, following an increase in global crude oil prices. However, the baht strengthened against the U.S. dollar again in early May, as risks in the global financial market heightened on the back of intensified trade tensions. This led to an appreciation of safe-haven currencies, particularly the yen and the baht. Additionally, capital flows returned to EMs following a more dovish stance of monetary policy communicated by the Fed and the ECB, a greater clarity on political developments in Thailand, as well as an increase in the weights of Thai equities and bonds in the MSCI and JP Morgan indices which induced foreign investors to increase holdings of Thai financial securities (Chart 2.13). Consequently, on 25 June 2019, the baht closed at 30.74 per U.S. dollar, up 3.3 percent from the end of the previous quarter.

[11/] Survey of credit conditions for the first quarter of 2019 and outlook for the second quarter of 2019.

The nominal effective exchange rate (NEER) index stood at 123.36 on 25 June 2019, an appreciation of 3.0 percent from the end of the previous quarter. The movement was in line with the baht appreciation against most currencies of Thailand's trading partners, with the exception of the yen. As the end of May 2019, the preliminary **real effective exchange rate (REER)** index rose about 1.5 percent from the end of the previous quarter, partly as Thailand's inflation rate rose at a faster pace than those of its trading partners. Going forward, exchange rates would likely remain volatile due to uncertainties surrounding trade tensions, monetary and fiscal policies of major advanced economies, geo-political risks, as well as political developments in Thailand.

Chart 2.12 The baht appreciated against the U.S. dollar

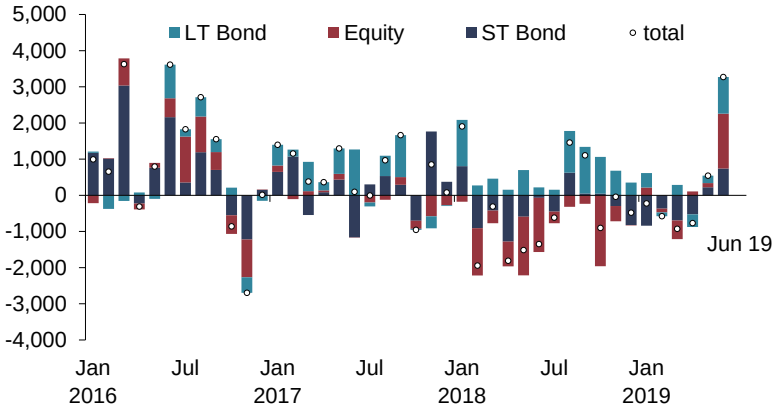
USDTHB, NEER, DXY



Source: Bank of Thailand and Reuters (data as of 25 June 2019)

Chart 2.13 Foreign portfolio investment in Thailand recorded net inflows

Million U.S. dollar



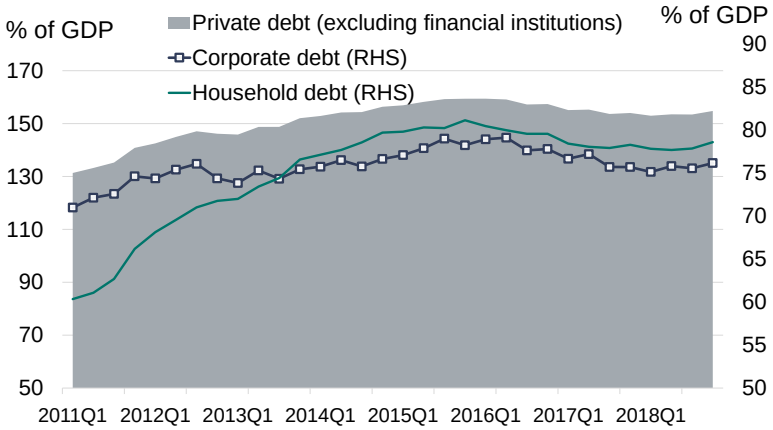
Source: Bloomberg and Thai BMA (Data as of 28 June 2019)

There remained pockets of risks in the Thai financial system that could pose vulnerabilities to financial stability in the future. These risks included: (1) an accumulation of household debt and debt serviceability of some household groups and small enterprises (2) search-for-yield behavior, particularly of saving cooperatives and large corporates, and (3) risks in the real estate sector.

Thailand's financial stability remained sound overall, particularly with regard to external stability, as indicated by the country's high level of international reserves and sustained current account surpluses, while the external debt to GDP ratio remained low. These factors combined to cushion the Thai economy against volatilities in global financial markets. Financial institutions maintained robust financial positions, as reflected in high levels of capital buffers to cushion against risks stemming from deterioration in credit quality. Nevertheless, there remained pockets of risks that warranted monitoring going forward. Such risks were as follows:

1. Leveraged household debt continued, while debt serviceability of some household groups and small enterprises continued to deteriorate. The ratio of household debt to GDP rose from the pervious quarter to 78.6 percent at the end of the fourth quarter of 2018 (Chart 2.14). Household debt was expected to continue rising due to the following factors: (1) a continued acceleration of auto loans by both commercial banks and leasing companies since the second quarter of 2017, as a result of sales promotions and more lenient credit standards on auto loans in the previous periods, and (2) the uptrend in mortgage loans prior to the implementation of the revised LTV ratio measures. This was observed in the expansion of new mortgage loans by financial institutions of 28.1 percent from the same period last year in the first quarter of 2019, an acceleration from the fourth quarter of 2018. Nevertheless, effects of such measures on debt accumulation behavior must be monitored going forward. However, persistently low interest rates induced households to incur new debts, which could weigh on future consumption and debt serviceability, and result in limited households' cushion against economic volatilities.

Chart 2.14 There was an increasing sign of household debt to GDP ratio
The ratio of private credit to GDP



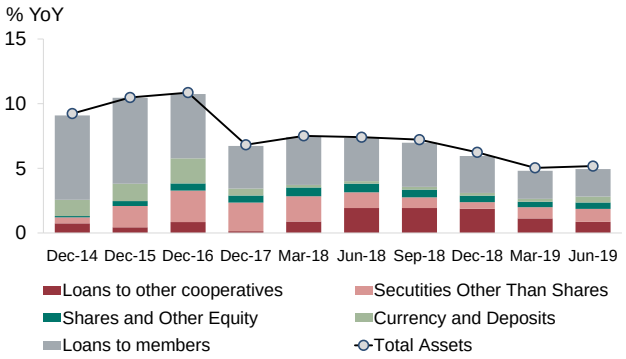
Source: Bank of Thailand

Overall credit quality remained stable, as indicated by the NPL ratio of commercial banks in the first quarter of 2019 which was stable at 2.9 percent relative to the previous quarter. However, debt serviceability of some loan types continued to deteriorate, particularly mortgage and auto loans whose NPL new-entry rates continued to trend up. Financial positions of some

small enterprises remained fragile, especially those in the construction and utilities sectors. These were reflected in operating profit margins (OPM) and sustained negative interest coverage ratio (ICR) of some small companies. [12/] As a result, the NPL ratio of commercial banks' SME loan portfolios remained high at 4.6 percent in the first quarter of 2019.

2. The continued search-for-yield behavior under the environment of persistently low interest rates could lead to underpricing of risks. Issues that warranted monitoring included, first, a continued search-for-higher-yield behavior of saving cooperatives, despite slower expansion in their assets and deposits relative to the previous periods (Chart 2.15). Systemic risk in the saving cooperatives system, both credit risk and liquidity risk, heightened from increased borrowings among saving cooperatives, especially if borrowers failed to repay loans. Second, some large corporates having significant connectedness with the financial system increasingly raised funds given a prolonged period of low interest rates. This was particularly done through the bond market, including perpetual bonds, which could lead investors on these instruments to underprice risks. Nevertheless, those corporates with high debts were in utilities and manufacturing sectors. Many of these businesses increasingly expanded foreign investments, particularly holding companies. The business structures of such businesses had become increasingly complicated, resulting in more difficult risk assessment and possibly underpricing of risks. Third, expansion of offshore investment through foreign investment funds (FIF) resumed growth since the beginning of this year, particularly FIFs that invested in debt instruments and foreign currency deposits. Concentration risks still warranted monitoring as FIF investments remained concentrated in only five major countries.

Chart 2.15 Assets of saving cooperatives expanded at a slower rate, but there remained pressures to search for higher yield
Contribution to growth of savings cooperatives' assets



Note: Saving cooperatives were subjected to tighter regulation by government since H2/2017
Source: Cooperative Auditing Department, calculations by Bank of Thailand

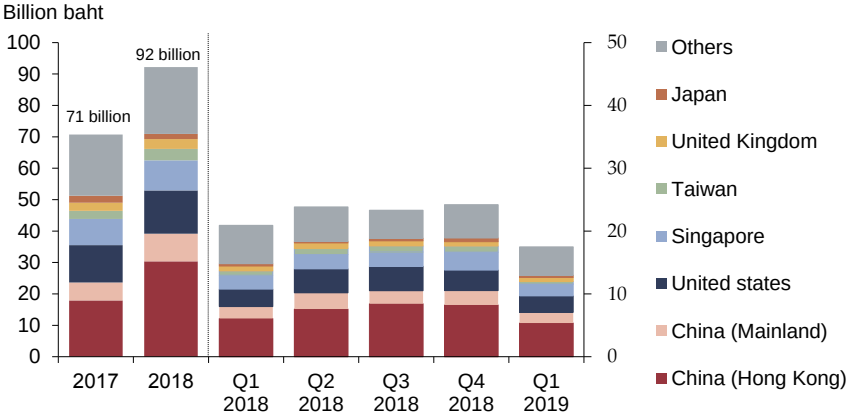
3. Risks in the real estate sector such as a decline in foreign demand, which might continue in response to a slowing outlook of the global economy. This was reflected in a 16.3 percent fall in the value of foreign funds transferred for condominium purchase in the first quarter of 2019 relative to the same period last year (Chart 2.16), which could lengthen time to sell existing excess supply. Nevertheless, risks in the real estate sector were partly addressed through the revised LTV ratio measures. This was observed in developers' adjustments by offering sales promotions for completed projects and delaying new project launches, and a largely stable housing price index after having continuously increased in the previous periods. In addition, financial institutions would be more cautious in expanding mortgage loans, as signs of moderation in speculative activities were observed as represented by a contraction in the number of loans for second and subsequent residential units acquired by the same purchasers. However, the increase in new loan approvals for low-rise houses and condominium purchases indicated acceleration in ownership transfers prior to the implementation of the revised LTV ratio measures (Chart 2.17). Nevertheless, there remained a

[12/] As reflected by the ICR and OPM of 25th-percentile small-sized listed companies which have been in the negative territory.

need to closely monitor developments in the real estate market after the revised LTV ratio measures became effective. These included credit standards of financial institutions and debt accumulation behavior of households.

Chart 2.16 In the first quarter of 2019, the value of funds transferred for condominium purchases by non-residents in terms of country of residence or nationality of the account owner declined from the same period last year

The value of funds transferred for condominium purchases by non-residents

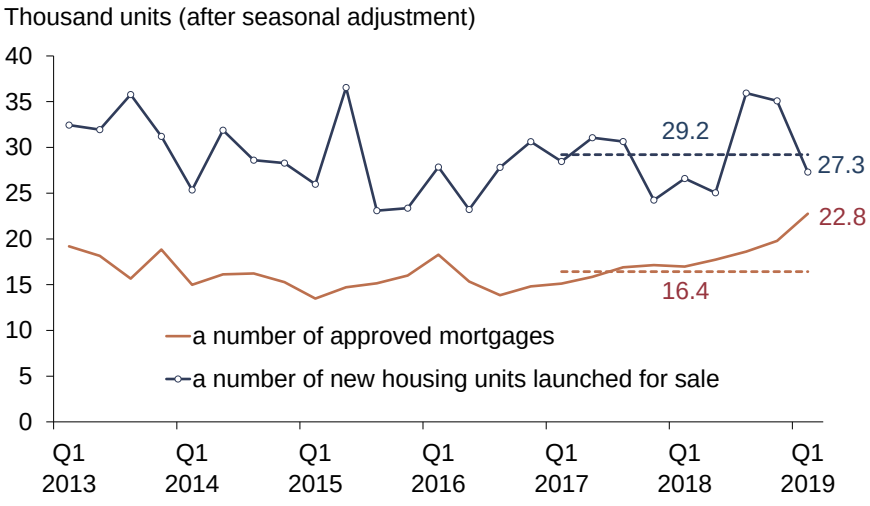


Note: The value of funds transferred for condominium purchases by non-residents is estimated from (1) the amount of foreign currencies sold for down payments or purchases of condominiums and (2) the amount of funds withdrawn from non-residents baht-denominated accounts for condominium purchases.

Source: Bank of Thailand.

Chart 2.17 A number of approved mortgages increased prior to the implementation of the revised LTV ratio measures, while developers focused more on selling completed units rather than launching new projects

A number of approved mortgages and new housing units launched for sale



Note: Dotted line represents average value for the past 6 years. (2013-2018)

Source: Bank of Thailand

The impact of climate variability on Thailand's agricultural sector

At the present time, global climate variability increases in intensity and recurrence. This is reflected by recurring phenomena of El Niño (drought) and La Niña (flood). Land surface temperature also rose continually and accelerated over the past decade. In particular, global temperature hit record highs for 4 consecutive years during 2015-2018. In 2018, global temperature was recorded as the fourth warmest year since global temperature records began in 1880. **Such climate variability poses adverse impact on global agricultural sectors, including Thailand, which rely heavily on the climate**.

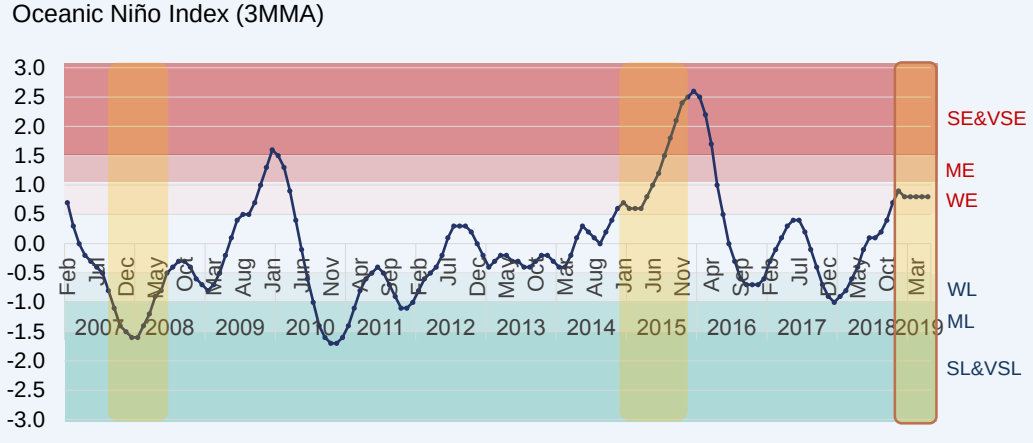
Over the recent years, intensified climate variability considerably devastated Thailand's agricultural output. The great flood in late 2011 destroyed in-season rice production in the central region's plain. Severe and persistent droughts during 2015-2016, regarded as the most severe drought crisis in the past 20 years, caused a contraction in Thailand's agricultural output in 2015 such as rice, fruits, and oil palm. This was the most severe damage to crop production in a decade. Meanwhile, crop prices also dropped, causing farm income to fall in 2015. These events demonstrated that more intensive climate variability poses direct impact on agricultural output and farm income. Hence, **climate conditions analysis is important for assessing the situation of Thailand's agricultural sector.**

1. Assessment of the current climate conditions

The following are two important indicators for assessing and monitoring climate conditions.

1.1 Oceanic Niño Index (ONI) - The ONI is an indicator of the probability of El Niño and La Niña phenomena. Currently, the index indicates **weak El Niño conditions** as the index has been positive in the range of 0.5-0.9 for five consecutive months (Chart 1). **Moreover, there is almost 60 percent probability of weak El Niño persisting until early 2020** (Chart 2). Although the current El Niño is less severe compared to the past events in 2015-2016, uncertainties remain high in the period ahead.

Chart 1 Oceanic Niño Index (ONI)

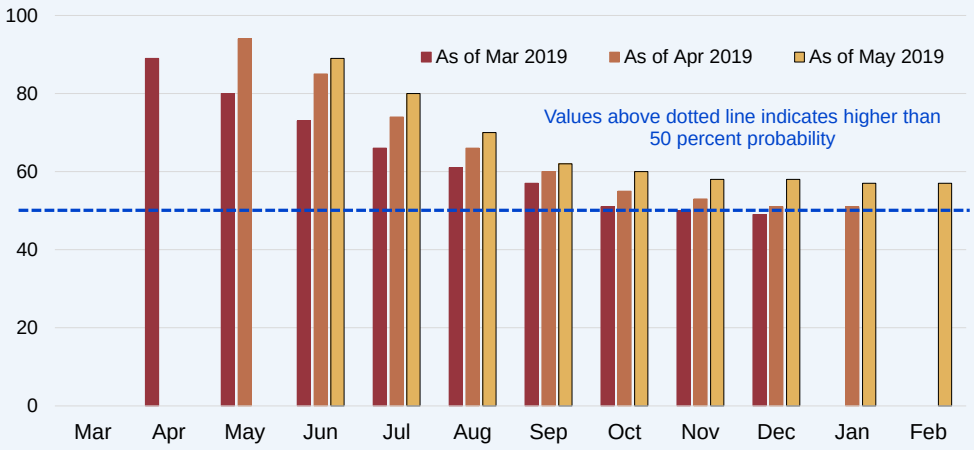


Red = El Niño (E) , Blue = La Niña (L), S = Severe , VS = Very Severe , M = Moderate & W = Weak

Note: El Niño event is identified if ONI is above 0.5 for 5 consecutive months.

Source: National Oceanic and Atmospheric Administration (NOAA)

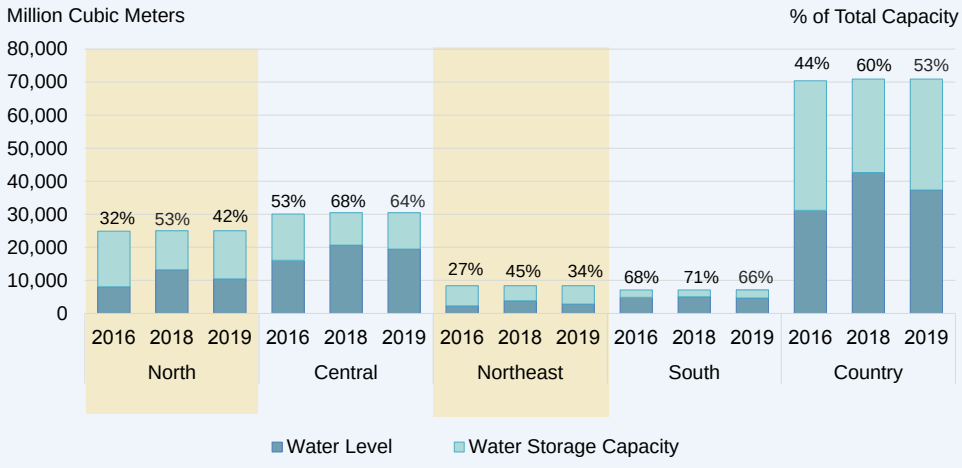
Chart 2 El Niño forecast probability



Source: International Research Institute for Climate and Society (IRI)

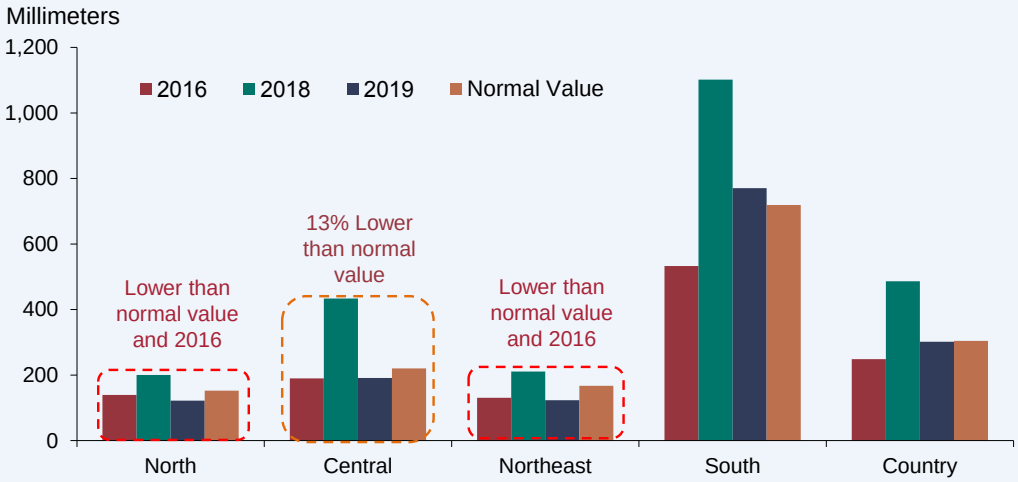
1.2 Indicators of Thailand's regional climate (1) *Reservoir water levels and accumulated precipitation.* Currently, the reservoir water levels reflected small impacts despite some signs of drought from delayed rainfall (Chart 3). (2) *Precipitation in the dry season* (November – April) had been close to the normal level (Chart 4). However, **the Northeast is more vulnerable to drought than other regions** as remaining water levels account for only one-third of reservoirs' capacity. Moreover, accumulated precipitation in dry season was below the normal level and the level in 2016, the year with a severe drought. In addition, since the size of most reservoirs in the Northeast are small, irrigated area accounts for only 11 percent of total area, which is the lowest among the regions (Chart 5).

Chart 3 Reservoir water level



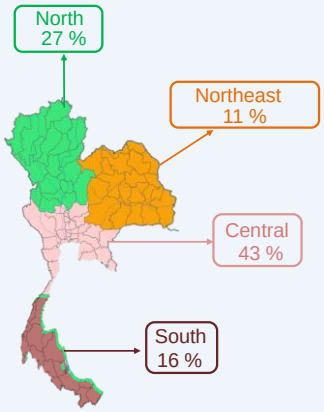
Sources: Royal Irrigation Department (data as of 4 June 2019)

Chart 4 Accumulated Precipitation in Dry Season by Regions (Nov – Apr)



Note: Normal value is an average of 30 years (1981 – 2010).
Source: Thai Meteorological Department, caluculated by BOT

Chart 5 Proportion of irrigated areas to total areas in 2017



Sources: Office of Agricultural Economics (OAE) and Bank of Thailand
Calculations by Bank of Thailand

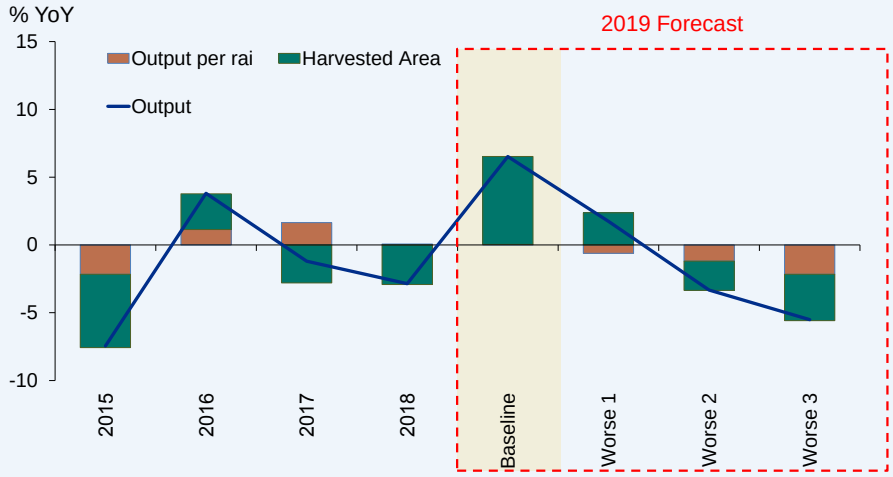
Going forward, although Thailand officially entered the wet season since late May, **weak El Niño conditions will likely persist for some time, leading to below-average precipitation**. Water levels in reservoirs are also lower than the previous year. Hence, there are high risks that the production of major crops cultivated during the wet season will be impinged.

2. Impact on agricultural output

Due to risks of climate uncertainties from May to October**,** which are the cultivating months for in-season rice as Thailand's major crop output**, this article aims to assess the impact of climate on the in-season rice output in 2019**. An econometric model is employed (1) to estimate output per rai given climate-related factors such as temperature, accumulated precipitation, dry season's reservoir discharge and cultivation area, as well as (2) to estimate harvest area using historical data.

The study reveals that under normal climate conditions, in-season rice output will expand by 6.5 percent. However**, if drought extend only across the Northeast (Scenario 1), in-season rice output will expand by only 1.7 percent.** This will be due to slower growth in the harvest area and the fall in output per rai following drought in some areas. **In case of more severe drought in the Northeast (Scenario 2) or of its spillover to the Northern and Central regions (Scenario 3), output will contract by 3.3 and 5.5 percent, respectively**. This will be owing to falls in both output per rai and harvest area. Nevertheless, the case study only illustrates drought impact on in-season rice. It does not consider impact on other crops, such as off-season rice and sugarcane, which may also be affected if drought expanded further (Chart 6).

Chart 6 Components contributing to in-season rice output growth under different drought scenarios



Sources: Office of Agricultural Economics (OAE) and Bank of Thailand, calculations by Bank of Thailand

Severe droughts will devastate agricultural output significantly. More intensifying climate variability including droughts and floods, as well as other types of risks such as plant and animal diseases, will pose direct impact on the well-being of farmers, who are one of the vulnerable groups with low income and high debt burden. Thus, **the government needs to formulate long-term resolution to tackle the problem through efficient water management together with appropriate and sufficient water allocation for consumption and cultivation.** In addition, the **government** needs to stand ready to handle such situation, to reduce risks for farmers who are the main affected party. This would promote their resilience against intensifying and recurring climate variability.

2.2 Outlook for the Thai Economy

Under the Committee's assessment, the Thai economy was expected to expand at a slower rate of 3.3 and 3.7 percent in 2019 and 2020, respectively, lower than the projection of 3.8 and 3.9 in the previous *Monetary Policy Report* (Table 2.2). The downward revision was mainly due to external demand, where growth of merchandise and services exports was projected to be lower than the previous assessment following a slowdown of trading partner economies and global trade volume. Moreover, private consumption and investment were expected to moderate from the previous year as merchandise exports contracted and confidence on future economic expansion somewhat declined. Public spending would continue to support economic growth in the period ahead albeit at a lesser extent than previously assessed, due to an expected delay in the enactment of the Annual Budget Expenditure Act, B.E. 2563 (A.D. 2020). The projection for headline inflation was in line with the previous assessment. Energy prices that were projected to be higher than expected would help offset lower-than-expected fresh food prices and core inflation. The latter was partly a result of demand-pull inflationary pressures that would likely decline in tandem with a softening economic growth outlook.

Table 2.2 Forecast summary

Percent	2018*	2019	2020
GDP growth	4.1	3.3 (3.8)	3.7 (3.9)
Headline inflation	1.1	1.0 (1.0)	1.0 (1.1)
Core inflation	0.7	0.7 (0.8)	0.9 (0.9)

Note: * Outturn

( ) Monetary Policy Report March 2019

Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

Summary of the key forecast assumptions

- Trading partner economies were projected to slow down throughout the forecast horizon due to (1) lower-than-expected growth outturns in the first quarter of 2019, particularly Asian economies, (2) slower-than-expected growth in several economies as reflected by economic indicators in the second quarter of 2019, and (3) uncertainties surrounding global trade outlook which affected production and investment in advanced economies and subsequently in regional countries.
- The federal funds rate was projected to be lower than the previous assessment, consistent with a more dovish stance of the Fed's monetary policy communication in order to cope with uncertainties pertaining to a softening outlook of the global economy. The Fed was expected to cut the policy rate once in the second half of 2019 with an additional cut in 2020, compared with the previous assessment of one policy rate hike in the second half of 2019, and maintaining the rate throughout 2020.
- Asian currencies (excluding the Chinese yuan) were expected to depreciate throughout the forecast horizon due to weaker-than-assessed outturns during the second quarter of 2019. In the period ahead, regional currencies would gradually and slowly appreciate, mainly on account of the outlook of the U.S. policy rate which would likely be more dovish.

- The Dubai crude oil price projection was maintained for 2019, but was revised downward for 2020 due to moderating demand for crude oil amid the global economic slowdown. However, tightening crude oil supply, due to a production cut according to the OPEC arrangement, was expected to support oil prices in the short term.
- Farm income was projected to expand at a faster pace than previously assessed on account of higher agricultural prices. This was particularly the case for rubber whose prices rose as global supply declined. Moreover, farm income was expected to further benefit from additional government measures through a budget increase in the project that aimed to help farmers grow other crops after rice cultivation.
- Public spending at current prices were revised downward throughout the forecast horizon, mainly due to public investment following the the delayed enactment of the Annual Budget Expenditure Act, B.E. 2563 (A.D. 2020). Moreover, infrastructure investment projects by some state-owned enterprises were postponed due to problems in accessing construction sites and changes in construction plans. Meanwhile, public consumption was expected to moderate slightly due to decelerating trend of personnel compensation expenditure.

Table: Summary of forecast assumptions

Annual percentage change	2018*	2019	2020
Dubai crude oil price (U.S. dollar per barrel)	69.6	65.9 (66.0)	65.0 (66.0)
Farm income (% YoY)	0.6	1.9 (1.0)	-0.4 (0.7)
Government consumption at current price (billion baht)[1/]	2,637	2,763 (2,771)	2,909 (2,918)
Public investment at current price (billion baht)[1/]	962	1,011 (1,034)	1,103 (1,122)
Fed funds rate (% at year end)	2.375	2.125 (2.625)	1.875 (2.625)
Trading partners' GDP growth (% YoY)[2/]	3.6	3.0 (3.2)	2.8 (3.2)
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)3/	153.8	156.8 (154.5)	156.9 (153.3)

Notes: [1/] Assumption includes spending on infrastructure investment plans

[2/] Weighted by each trading partner's share in Thailand's total exports

[3/] Increasing index represents depreciation, decreasing index represents appreciation

* Outturns

( ) Monetary Policy Report March 2019

The value of merchandise exports in 2019 would remain broadly unchanged from the previous year as a result of the slowdown in global economy and global trade volume.

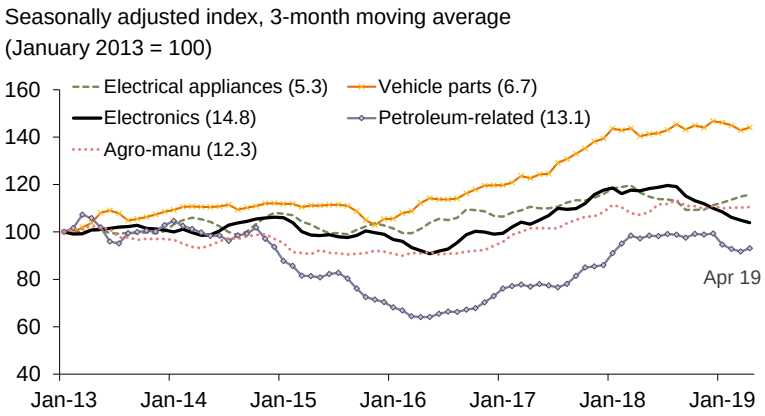
The value of merchandise exports in 2019 was expected to remain broadly unchanged from the previous year due to the impact from intensifying trade tensions and China's retaliatory measures, which could expand to non-tariff barriers (NTBs). Volume of Thai merchandise exports would be affected through following channels: (1) the deceleration of global trade volume, as reflected by the slowdown in the World Trade Outlook Indicator (WTOI)[13/] in the second quarter of 2019, and (2) a slowdown of external demand, as shown by a continued contraction in merchandise exports to China and several key trading partners in the recent period. Meanwhile, the benefits from redirected orders from China to Thailand could only partially offset a potential decline in exports (Chart 2.18 and 2.19). However, merchandise exports in the period ahead would

[13/] The World Trade Outlook Indicator (WTOI) is compiled by the World Trade Organisation (WTO).

be supported by the relocation of production base to Thailand of some businesses, including hard-disk drives and electrical appliances, as well as 5G technology-related infrastructure investment plans in many countries, which would promote global trade and investment. This would, in turn, benefit Thailand's exports of electronics parts going forward**. The Committee assessed that the value of merchandise exports in 2019 would remain broadly unchanged from the previous year and recover to register 4.3 percent growth in 2020, up from the previous forecast of 4.1 percent due to the lower base in 2019.**

Chart 2.18 Thai merchandise exports continued to contract in some product categories, except exports of auto parts and electrical appliances which were expected to grow from redirected orders and relocation of production base

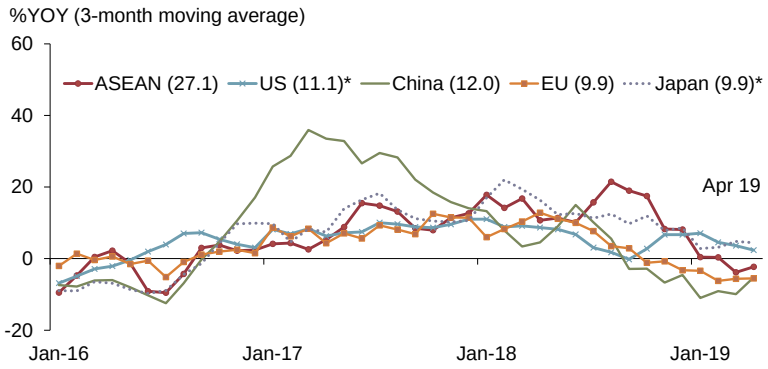
Value of merchandise exports, by product category



Note: Number in () denotes share to total exports in 2017
Source: Customs Department, calculations by Bank of Thailand

Chart 2.19 Merchandise exports contracted in almost all export destinations

Value of merchandise exports, by export destination



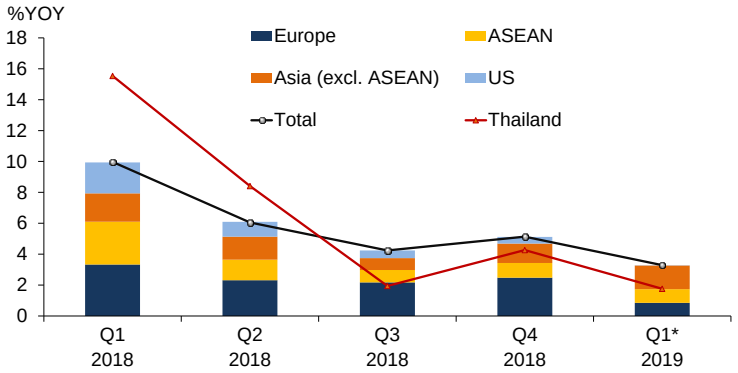
Note: Numbers in ( ) represent share in total exports in 2018
* Excluding exports of munitions for Cobra Gold Military Exercise
Source: Customs Department, calculations by Bank of Thailand

Exports of services were projected to decelerate amid the global economic slowdown.

Exports of services were projected to exhibit a slower growth than previously assessed due to a decreased number of foreign tourists and lower spending per head. In particular, foreign tourist figures and lower spending per trip declined during the first half of 2019, partly due to the global economic slowdown (Chart 2.20). However, government measures to promote tourism through the extended exemption of visa-on-arrival (VOA) fees until the end of October 2019 and a greater tourism promotion by both government and the private sector might partly help offset the impact. Therefore, the projected number of foreign tourists was revised down to 39.9 and 41.3 million in 2019 and 2020, respectively, lower than the previous projections of 40.4 and 42.0 million in the previous *Monetary Policy Report.*

Chart 2.20 A number of foreign tourists was expected to decline in many regions

Growth of foreign tourists, by tourist destination



Note: *calculated using a number of foreign tourist arrivals in 48 countries, accounting 50% of global tourists in 2018

Source: CEIC

The current account was projected to record 29.1 and 26.3 billion U.S. dollars surplus in 2019 and 2020, respectively. The estimates were revised down from the previous *Monetary Policy Report* due to a decline in the value of both merchandise and services exports. [14/]

Private consumption in 2019 was expected to moderate compared to the high growth last year, but would continue to expand.

Private consumption was projected to continue expanding across all product categories. Supporting factors included sales promotion continuously offered by businesses, rising farm household income mainly due to positive development of rubber prices, as well as government measures which partly helped support household consumption, such as government measures to boost spending through tax deduction and an increase in government transfers to support low-income earners through the social welfare card. Nonetheless, private consumption in the period ahead would be restrained by non-farm household income given signs of moderation in earnings, especially in export-related manufacturing sector.

[14/] Bank of Thailand has revised the data on tourism receipts since 2012 in accordance with the balance of payments concept. As a result, the revised surpluses during 2012-2018 have become smaller (further details in Najaree Nimitkamolchai and Chalisa Kalayanamitr (2019), "Revisions on Tourism Receipts in the Current Account", FAQ Issue 153, Bank of Thailand.

This could limit access to some types of loans for certain household groups as financial institutions would have to place greater emphasis on debt serviceability of borrowers. Moreover, consumer confidence among low-income households declined as elevated household debt pressured households to allocate part of their income for debt repayment.

Private investment was expected to continue expending despite at a lower rate than the previous assessment.

Private investment was expected to continue expending although the growth projection for 2019 was revised down from the previous estimate. This was due to a softer outlook of exports and a decline in investment confidence of businesses in the period ahead driven by the impact from trade protectionist measures. This was reflected in a postponement in investment of businesses, both investment for production efficiency improvement and capacity expansion. In 2020, private investment was expected to exhibit a higher-than-expected growth due to a greater clarity of some mega-investment projects, including the third phase of the Mabtapud Industrial Port and 5G technology-related infrastructure investment. Going forward, private investment would likely benefit from additional relocation of production base to Thailand, particularly Chinese industries affected by trade protectionist measures, and progress on infrastructure development in the Eastern Economic Corridor (EEC). The latter would attract greater private investment as reflected by the increasing number of investment promotion applications submitted to the Board of Investment (BOI) for investment projects within the EEC area.

Public spending was projected to support economic growth at a lesser extent than previously assessed.

Public spending was expected to register lower-than-expected growth. Public investment was revised downward for investment by the central government, due to an expected delayed in the enactment of the Annual Budget Expenditure Act, B.E. 2563 (A.D. 2020) as a result of a longer-than-usual period to form the new government. Also, some investment projects by state-owned enterprises were postponed due to contract reviews for the speed rail project between Bangkok and Nakhon Ratchasima, and problems in accessing contruction sites and changes in construction plans for the Purple Line mass rapid transit between Tao Poon-Rat Burana and the Missing Link along the Red Line.[15/] Meanwhile, public consumption was expected to slightly moderate due to decelerating trend of personnel compensation expenditure.

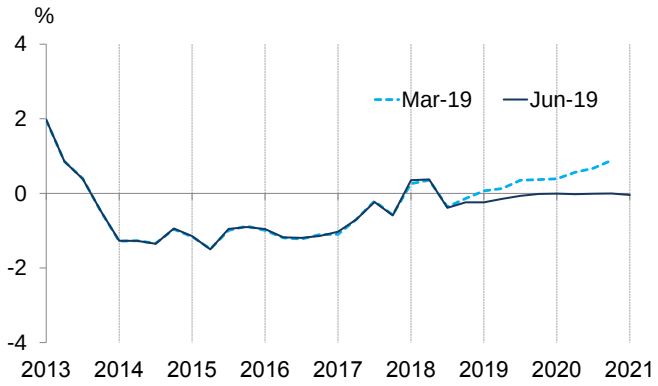
Projections of headline inflation were in line with the previous assessment. Core inflation was forecasted to be slightly lower than the previous estimate but would gradually trend up.

Headline inflation projection was expected to be in line with the previous projection in the previous *Monetary Policy Report*. Core inflation and fresh food prices were expected to be lower than previously anticipated due to lower-than-expected data outturns, and declining demand-pull inflationary pressures in the period ahead due to a softening economic growth outlook, as shown by the flattening output gap throughout the forecast horizon compared to the previous estimate (Chart 2.21). However, energy prices were projected to be higher than expected, especially in 2019, following higher-than-expected outturns of domestic retail oil

[15/] The Light-Red Line suburban railway from Bangsue, Phaya Thai, Makkasan to Hua Mak and the Dark-Red Line between Bangsue and Hua Lamphong.

prices. The increase was partly a result of delaying the assumption on a lower contribution to the energy fund to the third quarter of 2019, from previously anticipating a lower contribution in April. Meanwhile, core inflation in 2020 was projected to gradually rise in line with the outlook of domestic demand. Thus, **the Committee projected headline inflation to average 1.0 percent in 2019 and 2020, and core inflation to average 0.7 and 0.9 percent in 2019 and 2020, respectively.**

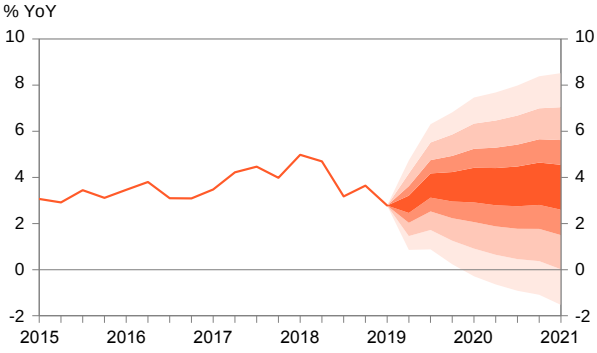
Chart 2.21 Output gap



Growth and inflation projections were still subject to downside risks.

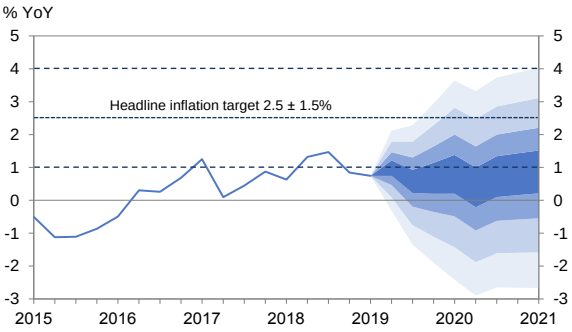
Under the Committee's assessment, **growth and inflation projections were subject to downside risks,** as reflected in the fan chart that tilted downward (Chart 2.22), despite, upside risks from additional government stimulus measures to support private spending. **Possibilities that the Thai economy would underperform the baseline projection** included (1) a larger-than-expected slowdown in global trade volume and impact on Thai merchandise exports in the case of intensified trade protectionist measures, (2) lower-than-expected trading parter economic growth, particularly in the event of no-deal Brexit, and due to consequences of Chinese economic and financial stability problems, and geopolitical risks, (3) a delay in the new government formation which could impact the continuation of government budget disbursement and policy implementation, and (4) a lower-than-expected growth of Chinese tourist figures if the Chinese government were to promote domestic tourism. However, there were **possibilities that the Thai economy would outperform the baseline projection** owing to (1) higher-than-expected growth of Thailand's trading partner economies, should there be additional government stimulus measures or better-than-expected improvements on trade tensions, and (2) higher-than-expected domestic demand as a result of (2.1) a sooner-than-expected implementation of government infrastructure investment projects, PPP projects and private investment, and (2.2) a larger-than-expected contribution from government stimulus measures if the new government could accelerate disbursements or release additional measures to stimulate private spending. Meanwhile, **risks to the forecasts of headline and core inflation were expected to tilt downward in line with risks** to growth projections and assumptions on global crude oil price (Chart 2.23 and Chart 2.24).

Chart 2.22 Growth forecast



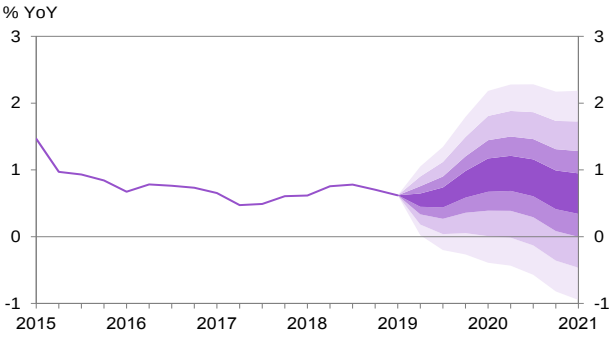
Note: Fan chart covers 90% of the probability distribution

Chart 2.23 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.24 Core inflation forecast



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2018*	2019	2020
GDP growth	4.1	3.3 (3.8)	3.7 (3.9)
Domestic demand	3.9	3.5 (3.9)	3.9 (3.9)
Private consumption	4.6	3.8 (3.9)	3.4 (3.7)
Private investment	3.9	3.8 (4.4)	5.5 (5.0)
Government consumption	1.8	2.2 (2.3)	2.6 (2.6)
Public investment	3.3	3.8 (6.1)	7.2 (6.6)
Exports of goods and services	4.2	0.3 (3.1)	3.9 (3.7)
imports of goods and services	8.6	0.7 (2.7)	4.5 (4.2)
Current account (billion, U.S. dollars)	32.4	29.1 (34.5)	26.3 (31.5)
Value of merchandise exports	7.5	0.0 (3.0)	4.3 (4.1)
Value of merchandise imports	13.7	-0.3 (3.1)	4.8 (4.8)
Number of foreign tourists (million person)	38.3	39.9 (40.4)	41.3 (42.0)

Note: *Outturns, which may be revised according to changes in statistical compilation methods

( ) Monetary Policy Report March 2019

3. Monetary Policy Decision

The Committee assessed that the Thai economy would expand at a slower pace than previously assessed amid external and internal uncertainties in the period ahead. Accommodative monetary policy thus remained appropriate for fostering sustainable economic growth while preserving price stability. Meanwhile, safeguarding financial stability would remain necessary and could be done a combination of tools.

The Committee weighed various factors in determining the most appropriate course of monetary policy, placing great emphasis on the strength and continuation of economic growth, development of headline inflation, and preserving financial stability. Details on the assessment of latest developments and outlook are as follows.

1. Economic growth The Thai economy was expected to expand at a slower pace than the previous assessment due to merchandise exports outlook which exhibited significantly lower-than-expected growth. This was owing to trading partner economies' and global trade slowdown following intensified trade tensions. Moreover, exports of services were expected to moderate, especially tourist figures and Chinese tourists' spending per head. With regard to domestic demand, private consumption was projected to continue expanding, but would be pressured by the purchasing power of non-farm households, particularly those in export-related manufacturing sectors, as well as by elevated household debt. Meanwhile, private investment would likely grow at a lower rate than the previous assessment, as some investors delayed their investment in response to softer exports outlook and lower investment confidence. However, private investment in 2020 was expected to continue expanding thanks to the relocation of production base to Thailand due to the impact of trade protectionist measures, as well as progress in the EEC investment projects. Public expenditure would help drive the economy to a lesser extent than previously assessed due to an expected delay in the enactment of the 2020 Annual Budget Expenditure Act as well as the postponement of some state-owned enterprise investment. The Committee assessed that there remained risks of the Thai economy underperforming baseline projection due to (1) potentially intensifying trade tensions, which could affect domestic demand, as well as (2) the ability of the government to advance important policies and maintain the continuity of public spending.

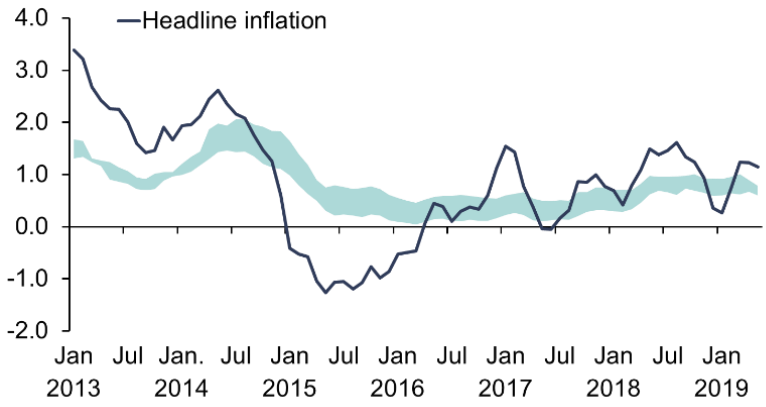
2. Preserving financial stability The Committee assessed that there remained certain pockets of risks that might pose vulnerabilities to financial stability in the future. Such risks included **(1) high and rising household leverage,** particularly auto leasing and mortgage loans. Credit quality of auto leasing loans continued to deteriorate, partly due to more lenient credit underwriting standards over the previous periods. This was partly a result of heightened competition from increasing participation of non-bank financial institutions. Also, farm income and household income in the provinces decelerated significantly in recent periods. Meanwhile, credit quality of SMEs was similarly expected to deteriorate, especially in the trading sector where the new-entry NPL rate remained high. In addition, a prolonged low interest rate environment induced households to create new debt, which could weigh on consumption and affect debt serviceability in the future. **(2) The continued search-for-yield behavior given a prolonged low interest rate environment could lead to underpricing of risks.** For example, saving cooperatives continued to search for higher returns as reflected in the high growth of

their assets and deposits, although somewhat decelerated. In addition, large corporates continued to raise funds given a prolonged low interest rate environment. These corporates had high interconnectedness in the financial system, especially through bond issuances. **(3) Risks in the real estate sector remained,** including an oversupply of condominiums in certain areas due to likely declines in the foreign demand for Thai condominiums from global economic slowdown. Nevertheless, there was an acceleration in ownership transfers before the implementation of the revised LTV measures as reflected by increased new loan approvals for both low-rise houses and condominiums. Meanwhile, real estate developers adjusted by launching sale promotions on finished projects and delayed launching new projects. However, there was a need to closely monitor adjustments in the real estate market and in mortgage loans after the revised LTV measures' effective date in April 2019.

3. Inflation development Headline inflation would be largely unchanged from the previous projection. Core inflation and fresh food prices would likely decline in line with data outturns and moderating demand-pull inflationary pressures. This was reflected by various indicators including trimmed mean and principal component indicators. (Chart 3.1) Moreover, structural changes such as intensified price competition and cost-reduction technology partly caused inflation to rise at a slower pace than in the past. Nevertheless, rising energy prices would help offset possible declines in core inflation and fresh food prices. Meanwhile, **short-term inflation expectations** (1-year ahead) were largely unchanged from the previous quarter. (Chart 3.2)

Chart 3.1 Demand-pull inflationary pressures decreased

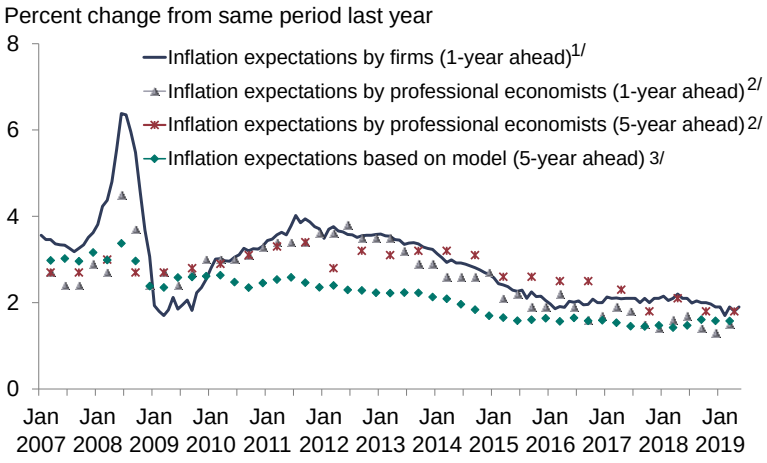
Percent change from same period last year



Note: The field shows the highest and lowest outcomes among different measures of underlying inflation. The measures included are 1. **Asymmetric trim** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute price movements across categories of goods and services) and 3. **Core inflation excluding rents and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 3.2 Inflation expectations remained stable from the previous periods



Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

There remained a need to employ a combination of tools to reduce financial stability risks in the period ahead.

The Committee discussed measures to address and prevent financial stability risks. The Committee viewed that safeguarding financial stability in the period ahead would remain necessary and could be done through a combination of tools, including the appropriate policy interest rate, macroprudential measures, and microprudential measures. This was because the sole use of macroprudential and microprudential measures in recent periods could only address certain risks. Therefore, the Committee deemed it necessary to continue monitoring the impacts of the implemented macroprudential measures. Regarding the current context of monetary policy, more emphasis was placed on preserving financial stability. The Committee discussed a framework for incorporating financial stability consideration in monetary policy formulation in a more systematic manner. The Committed developed comprehensive financial stability indicators, including a financial cycle indicator that was related to downside risks of economic growth, in order to analyze basic tradeoffs between macroeconomic and financial stability objectives. Moreover, the Committee viewed that an appropriate mix of monetary policy tools was still necessary in the period ahead to curb financial stability. In conjunction with that, a further examination of additional measures was required to prevent systemic risks more effectively.

The Committee voted unanimously to maintain the policy rate at 1.75% in order to foster accommodative financial conditions. This would foster a continuation of economic growth and was appropriate given the inflation target, while also not posing vulnerabilities to financial stability.

The Committee voted unanimously to keep the policy unchanged at 1.75% at the meetings on 8 May and 26 June 2019. The Committee viewed that the current accommodative monetary policy stance contributed to the continuation of economic growth and was appropriate given the inflation target. Moreover, against the backdrop of heightened uncertainties surrounding global economic conditions and domestic factors, the Committee thus deemed it appropriate to keep the policy rate unchanged. However, the Committee expressed concerns

over a somewhat rapid appreciation of the baht which might not be consistent with economic fundamentals and could negatively affect economic growth going forward. Given moderating economic prospects, current economic data from various sectors reflected that the Thai economy would be more sensitive to currency appreciation. The labor market showed clearer signs of vulnerabilities, especially in the export-related manufacturing and tourism sectors. Therefore, developments of exchange rates, capital flows, as well as impacts on the economy through various channels warranted close monitoring. Along with that, it was necessary to prepare short-term capital inflow management measures ready to be implemented at an appropriate time, as well as to continue relaxing more capital outflow regulations to encourage a greater flow of outward portfolio investment by residents. In the period ahead, there remained a need to address financial stability risks through a combination of tools, including the appropriate policy rate as well as microprudential and macroprudential measures which would need to place greater emphasis on debt serviceability of borrowers. In addition, the Committee deemed it necessary to examine further appropriate macroprudential measures to prevent systemic risks more effectively.

Looking ahead, there remained high uncertainties from both external and internal factors. The current accommodative monetary policy would, therefore, remain appropriate. However, the Committee would closely monitor developments of the outlook for growth, inflation, financial stability, and exchange rate, together with associated risks, especially impacts of trade tensions, in deliberating appropriate monetary policy in the period ahead.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2017	2018	2018				2019
			Q1	Q2	Q3	Q4	Q1
GDP growth	4.0	4.1	5.0	4.7	3.2	3.6	2.8
Production							
Agriculture	3.7	5.1	6.8	10.5	3.2	0.7	0.9
Non-agriculture	4.1	4.0	4.8	4.2	3.1	4.0	3.0
Manufacturing	3.0	3.0	3.7	3.1	1.5	3.5	0.6
Construction	-2.8	2.7	1.2	1.9	4.5	3.4	3.0
Wholesales and retail trade	7.0	7.3	7.0	7.4	7.5	7.5	6.8
Accommodation and Food Service	10.6	7.9	13.1	8.8	4.1	5.3	4.9
Transport and storage	8.2	5.7	7.1	5.4	4.5	5.4	3.4
Information and Communication	2.7	6.8	5.2	7.7	7.4	6.9	6.8
Financial intermediation	5.4	3.3	3.5	4.6	3.1	1.8	2.4
Real estate, renting, and business activities	6.5	5.2	6.0	4.0	5.9	4.9	4.7
Expenditure							
Domestic demand	2.1	3.9	3.3	3.6	4.3	4.3	4.0
Private consumption	3.0	4.6	3.8	4.1	5.2	5.4	4.6
Private investment	2.9	3.9	3.1	3.1	3.8	5.5	4.4
Government consumption	0.1	1.8	1.8	2.3	1.9	1.4	3.3
Public investment	-1.2	3.3	4.0	4.9	4.2	-0.1	-0.1
Imports of goods and services	6.2	8.6	9.1	8.8	11.0	5.7	-0.2
imports of goods	7.4	8.1	10.4	7.9	9.9	4.5	-2.6
imports of services	1.3	10.7	3.9	12.8	16.1	10.4	10.3
Exports of goods and services	5.4	4.2	8.0	9.6	-0.9	0.7	-4.9
exports of goods	5.7	4.1	7.2	9.5	-0.5	0.8	-5.4
exports of services	4.6	4.4	9.9	10.3	-2.2	0.0	-3.6
Trade balance (billion, U.S. dollars)	32.6	22.4	7.4	6.4	3.9	4.6	6.5
Current account (billion, U.S. dollars)*	44.1	32.4	14.6	6.3	4.5	7.0	12.5
Financial account (billion, U.S. dollars)	-12.4	-20.8	-3.7	-9.5	-4.0	-3.6	-5.9
International reserves (billion, U.S. dollars)	202.6	205.6	215.6	206.8	204.5	205.6	212.2
Unemployment rate (%)	1.2	1.1	1.2	1.1	1.0	0.9	0.9
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	1.2	1.0	1.0	1.0	0.9

Note: *Outturns, which are subject to revisions according to changes in statistical compilation methods

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2017	2018	2018				2019		
			Q1	Q2	Q3	Q4	Q1	Apr	May
1. Financial market sector									
Bond market									
Bond spread (10 years - 2 years)	1.1	0.7	1.2	1.1	0.9	0.7	0.7	0.7	0.6
Equity market									
SET index (end of period)	1,753.7	1,563.9	1,776.3	1,595.6	1,756.4	1,563.9	1,638.7	n.a.	n.a.
Actual volatility of SET index[1/]	6.5	12.1	9.4	12.3	11.6	14.0		n.a.	n.a.
Price to Earnings ratio (P/E ratio) (times)	19.1	14.8	18.3	16.2	17.3	14.8	17.1	n.a.	n.a.
Exchange rate market									
Actual volatility of Thai baht (%annualized)[2/]	4.4	3.3	4.6	4.4	4.5	4.7	4.6	3.6	4.4
Nominal Effective Exchange Rate (NEER)	110.6	115.6	114.8	115.2	115.2	117.0	119.9	120.5	122.6
Real Effective Exchange Rate (REER)	103.7	107.2	106.3	107.1	107.1	108.5	110.2	110.0	111.3
2. Financial institution sector[3/]									
Minimum Lending Rate (MLR)[4/]	6.28	6.28	6.28	6.28	6.28	6.28	6.28	6.28	6.28
12-month fixed deposit rate[4/]	1.37	1.37	1.37	1.37	1.37	1.37	1.42	1.42	1.46
Capital adequacy									
Capital funds / Risk-weighted asset (%)	18.2	18.3	18.1	17.9	18.4	18.3	18.2	18.5	n.a.
Earning and profitability									
Net profit (billion, Thai baht)	187.1	207.2	50.2	56.4	51.1	49.5	57.1	n.a.	n.a.
Return on assets (ROA) (times)	1.1	1.1	1.1	1.2	1.2	1.1	1.2	n.a.	n.a.
Liquidity									
Loan to Deposit and B/E (%)	96.1	98.2	95.0	96.8	98.2	98.2	96.8	96.1	n.a.
3. Household sector									
Household debt to GDP (%)	78.4	78.6	77.8	77.7	77.9	78.6	n.a.	n.a.	n.a.
Financial assets to debt (times)	2.6	n.a.	2.7	2.6	2.5	n.a.	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Consumer loans	2.7	2.7	2.8	2.7	2.7	2.7	2.8	n.a.	n.a.
Housing loans	3.2	3.2	3.4	3.4	3.4	3.2	3.4	n.a.	n.a.
Auto leasing	1.6	1.7	1.5	1.5	1.6	1.7	1.7	n.a.	n.a.
Credit cards	2.6	2.3	3.2	2.4	2.5	2.3	2.7	n.a.	n.a.
Other personal loans	2.5	2.5	2.7	2.5	2.5	2.5	2.6	n.a.	n.a.
4. Non-financial corporate sector[5/]									
Operating profit margin (OPM) (%)	8.0	7.5	7.9	7.8	8.0	6.5	7.7	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.8	0.7	0.7	n.a.	n.a.
Interest coverage ratio (ICR) (times)	6.5	6.4	7.1	6.5	6.6	5.4	5.1	n.a.	n.a.
Current ratio (times)	1.7	1.6	1.7	1.7	1.6	1.6	1.6	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Large businesses	1.8	1.7	1.7	1.7	1.5	1.7	1.5	n.a.	n.a.
SMEs	4.4	4.5	4.5	4.5	4.7	4.5	4.6	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 5 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2017	2018	2018				2019		
			Q1	Q2	Q3	Q4	Q1	Apr	May
5. Real estate sector									
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)									
Total	65,124	73,345	15,011	17,917	18,803	21,614	19,710	10,129	4,131
Single-detached and semi-detached houses	14,517	15,906	3,527	3,825	3,947	4,607	4,446	2,367	867
Townhouses and commercial buildings	21,469	25,039	5,166	6,055	6,324	7,494	7,290	3,699	1,384
Condominiums	29,138	32,400	6,318	8,037	8,532	9,513	7,974	4,063	1,880
Number of new housing units launched for sale (Bangkok and Vicinity) (units)									
Total	110,575	114,477	26,829	19,836	42,451	35,861	27,331	10,839	4,528
Single-detached and semi-detached houses	19,433	14,280	4,223	2,790	6,248	5,050	3,644	2,490	449
Townhouses and commercial buildings	32,792	36,571	6,657	6,548	8,759	10,385	4,233	2,100	1,544
Condominiums	58,350	63,626	15,949	10,498	27,444	20,426	19,454	6,249	2,535
Housing price index (2009 = 100)									
Single-detached houses (including land)	130.9	138.8	137.7	137.8	139.8	139.9	141.9	141.9	142.3
Townhouses (including land)	141.1	149.9	145.9	149.5	151.7	152.3	156.4	156.4	156.3
Condominiums	169.6	180.9	179.8	176.7	180.4	186.6	180.2	180.2	176.5
Land	171.7	175.8	175.7	177.2	172.7	177.8	175.6	175.6	173.6
6. Fiscal sector									
Public debt to GDP (%)	40.7	41.2	41.2	41.0	42.1	41.9	41.9	42.1	n.a.
7. External sector									
Current account balance to GDP (%)[6/]	9.7	6.4	11.4	5.0	3.7	5.4	9.3	n.a.	n.a.
External debt to GDP (%)[7/]	36.7	35.1	36.6	35.1	35.3	35.1	34.9	n.a.	n.a.
External debt (billion, U.S. dollars)	155.2	160.9	157.9	154.3	158.2	160.9	163.4	160.5	163.2
Short-term (%)	44.3	38.6	43.1	43.0	41.4	38.6	38.2	37.3	37.6
Long-term (%)	55.7	61.4	56.9	57.0	58.6	61.4	61.8	62.7	62.4
International reserves / Short-term external debt (times)	2.9	3.3	3.2	3.1	3.1	3.3	3.4	3.5	3.4

Note:

[6/] Current account / Nominal GDP at the same quarter, based on current account data for 2018 as published on 29 June 2019

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2019			2020				2021
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 9	0	0	0	1	1	2	3	4
8-9	0	0	1	2	2	3	3	4
7-8	0	2	3	4	5	5	6	6
6-7	0	5	7	8	8	8	9	8
5-6	3	13	13	13	12	12	12	11
4-5	13	21	18	16	15	14	13	13
3-4	29	24	20	17	16	15	14	13
2-3	31	19	17	15	14	13	12	12
1-2	18	10	11	11	11	11	10	10
0-1	5	4	6	7	7	8	7	8
(-1)-0	1	1	3	4	4	5	5	5
(-2)-(-1)	0	0	1	2	2	3	3	3
(-3)-(-2)	0	0	0	1	1	1	1	2
< (-3)	0	0	0	0	1	1	1	2

Table: Probability distribution of headline inflation forecast

Percent	2019			2020				2021
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 4.0	0	0	1	3	2	3	4	5
3.5-4.0	0	0	1	3	2	3	3	3
3.0-3.5	0	1	2	4	3	4	5	5
2.5-3.0	1	2	5	6	5	6	6	7
2.0-2.5	5	5	8	8	7	8	8	8
1.5-2.0	16	10	11	10	8	9	9	9
1.0-1.5	25	15	13	11	10	10	10	10
0.5-1.0	24	18	13	11	10	10	10	10
0.0-0.5	17	16	12	10	10	10	9	9
(-0.5)-0.0	8	13	11	9	9	9	8	8
(-1.0)-(0.5)	3	9	8	8	8	7	7	7
(-1.5)-(1.0)	1	5	6	6	7	6	6	6
(-2.0)-(-1.5)	0	3	4	5	6	5	5	4
< -2.0	0	2	5	9	12	10	10	10

Table: Probability distribution of core inflation forecast

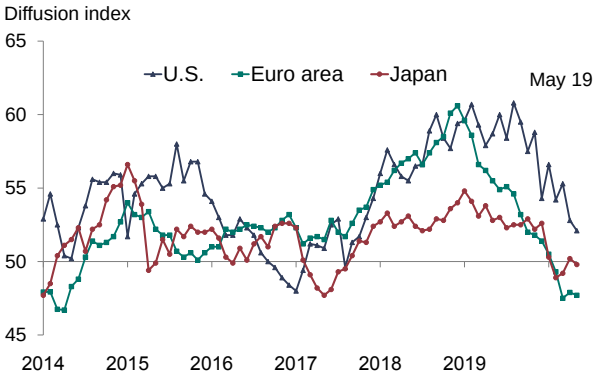
Percent	2019			2020				2021
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 4.0	0	0	0	0	0	0	0	0
3.5-4.0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	0	0	0
2.5-3.0	0	0	0	1	2	2	1	1
2.0-2.5	0	0	1	5	6	6	4	5
1.5-2.0	0	2	8	13	14	13	11	10
1.0-1.5	5	14	23	22	22	21	18	17
0.5-1.0	47	38	30	24	23	22	21	20
0.0-0.5	42	33	23	18	17	18	19	19
(-0.5)-0.0	6	12	10	10	10	11	13	14
(-1.0)-(0.5)	0	2	3	4	4	5	7	8
(-1.5)-(1.0)	0	0	1	1	1	2	3	4
(-2.0)-(-1.5)	0	0	0	0	0	1	1	1
< -2.0	0	0	0	0	0	0	0	1

4.2 Chart pack

Global Economy

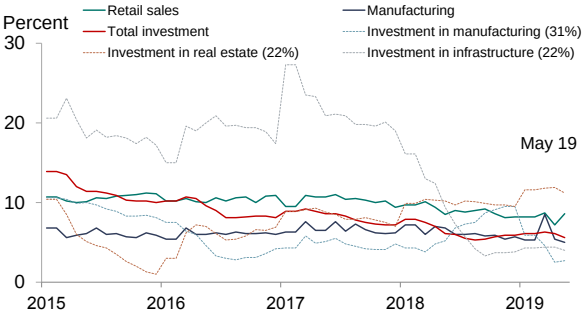
Major advanced economies were expected to grow at a slower rate, mainly on account of exports and private investment that were affected by the trade protectionist measures between the U.S. and China, despite higher-than-expected economic data outturns of several economies in the first quarter of 2019. However, robust labor market and government stimulus measures in many countries would be supporting factors for growth of Thailand's trading partner economies in the period ahead.

Manufacturing Purchasing Manager Index



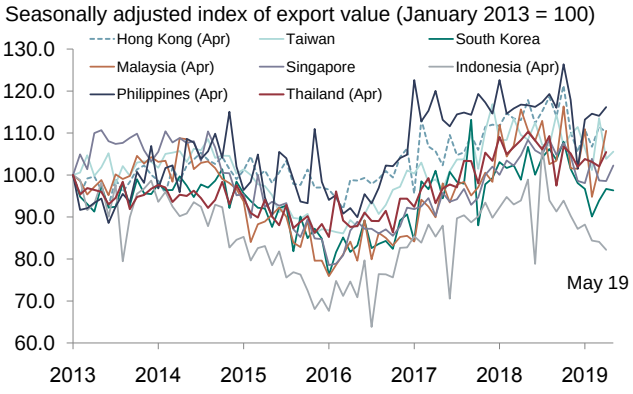
Sources: Bloomberg and Eurostat

China's economic indicators (Change from the same period last year)



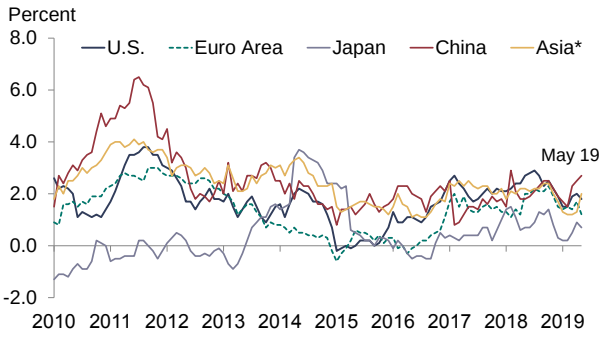
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners



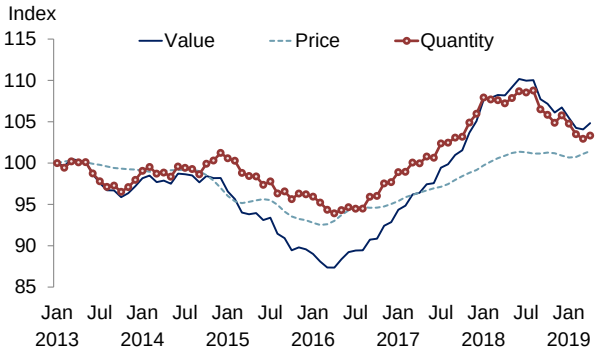
Note: *Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

Thai Economy

The Thai economy expanded at a slower pace, as merchandise exports were mainly affected by a slowdown of trading partner economies and trade protectionist measures. However, domestic demand continued to support growth, as private consumption and investment continued to expand. Meanwhile, government expenditure expanded on account of both public consumption and investment.

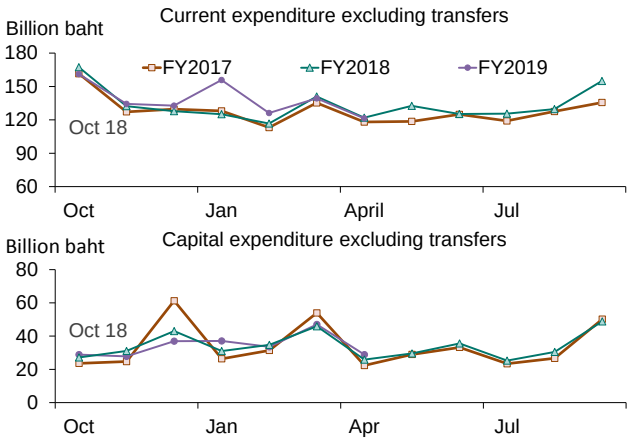
Thai exports (excluding gold): value, price and quantity

(3-month moving average, seasonally adjusted; January 2013 = 100)



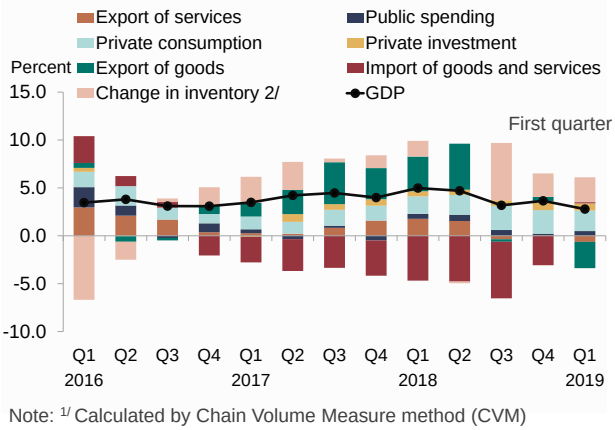
Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

Public spending by central government



Source: Bureau of Budget, Fiscal Policy Office

Contribution to Thailand's GDP growth[1/]

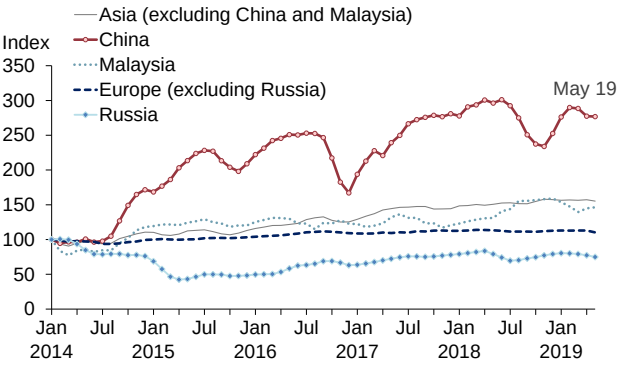


Note: [1/] Calculated by Chain Volume Measure method (CVM)
[2/] Change in inventory and statistical discrepancy

Source: Office of National Economic and Social Development Council, calculations by the Bank of Thailand

Index of foreign tourists classified by nationality

(three-month moving average, seasonally adjusted; January 2014 = 100)

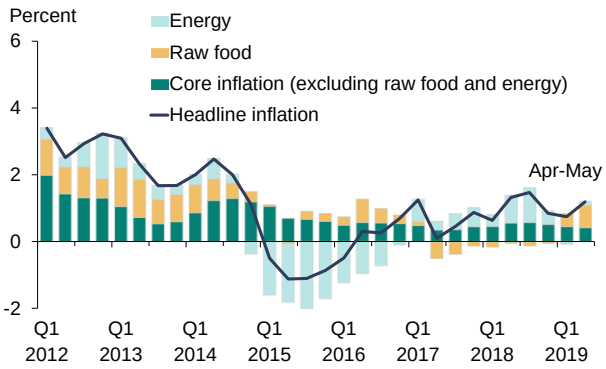


Source: Department of Tourism

Inflation

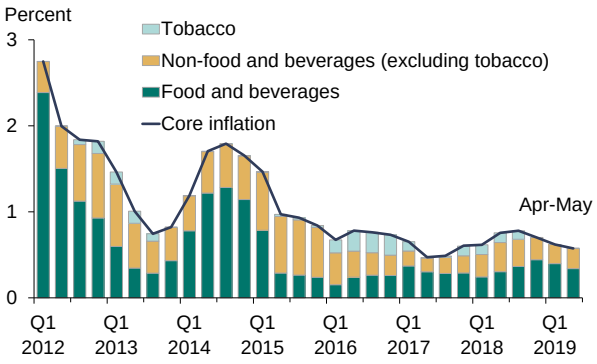
Headline inflation edged up, particularly in the fresh food category. Meanwhile, core inflation declined slightly mainly in line with core inflation in the food category. Nevertheless, prices of most goods and services slowly increased despite growing domestic demand, since structural changes led to a moderate increase in core inflation in the non-food categories in the recent periods.

Contribution to headline inflation



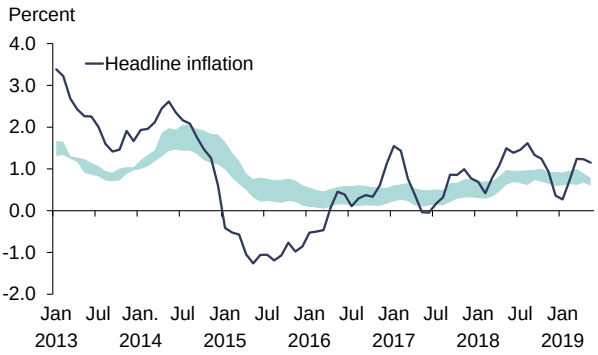
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

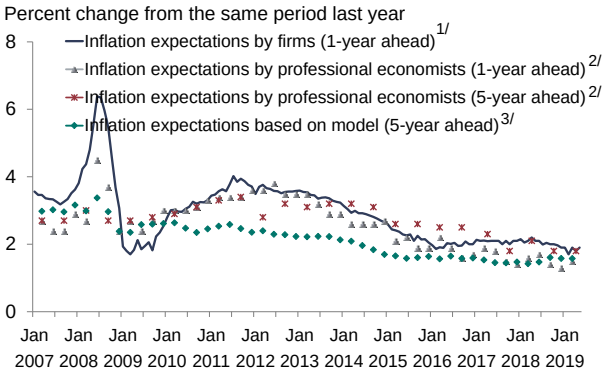
Underlying inflation indicators



Note: The field shows the highest and lowest outcomes among different measures of underlying inflation. The measures included are 1. **Trimmed mean** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute price movements across categories of goods and services) and 3. **Core inflation excluding rents and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

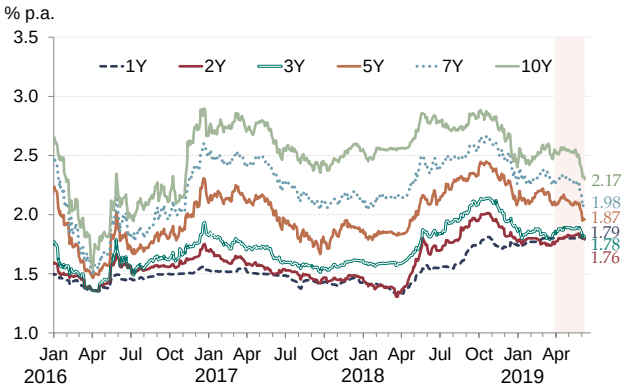


Source: 1/ Business Sentiment Survey (BSI) by the Bank of Thailand
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model using bond yield and macroeconomic data

Financial conditions

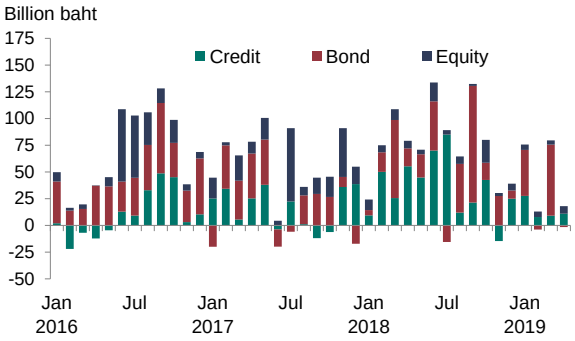
Short-term money market rates were stable around the policy rate. Short-term government bond yields gradually rose. Meanwhile, medium- and long-term government bond yields declined in line with the U.S. government bond yields. Since the end of the previous quarter, the baht appreciated against the U.S. dollar, following a more dovish stance of monetary policy communicated by the Fed and a greater clarity pertaining to political developments in Thailand, which induced foreign investors to increase holdings of Thai securities. Financial conditions remained accommodative, which allowed corporate financing to continue.

Government bond yields



Source: Thai Bond Market Association (Thai BMA) (data as of 25 June 2019)

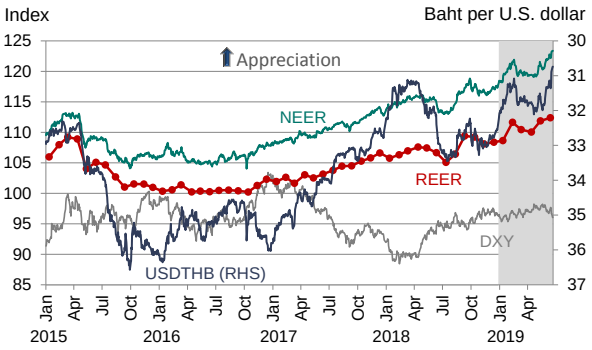
Total corporate financing by instrument*



Note: *Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding those issued by commercial banks, and newly issued equities.

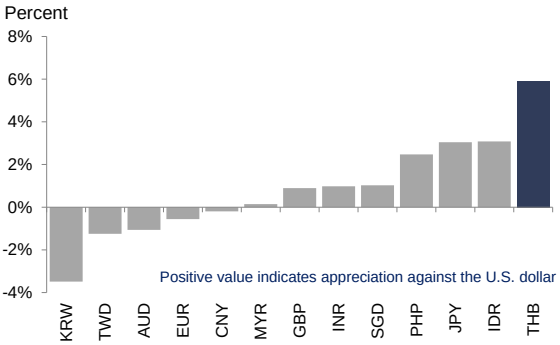
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and the Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of 25 June 2019)

Currency movements vis-a-vis the U.S. dollar (25 June 2019 compared to 28 December 18)

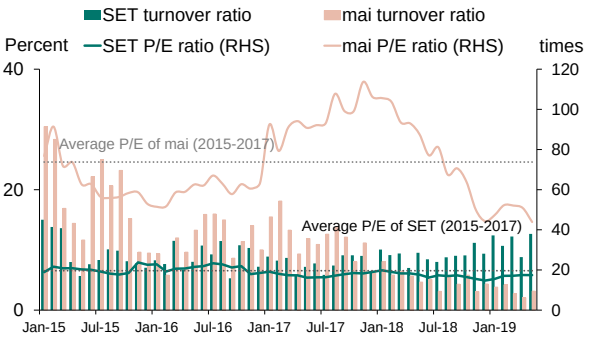


Source: Bank of Thailand and Reuters (data as of 25 June 2019)

Stability: financial markets

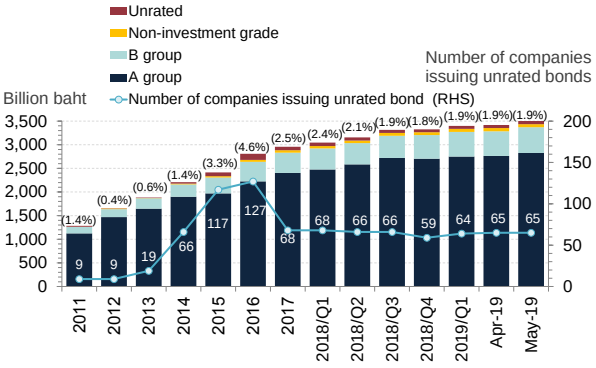
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand (SET) stayed below its historical average, while that of the Market for Alternative Investment (mai) decreased. The share of unrated bond issuance remained stable.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of February 2019)

Corporate bonds outstanding

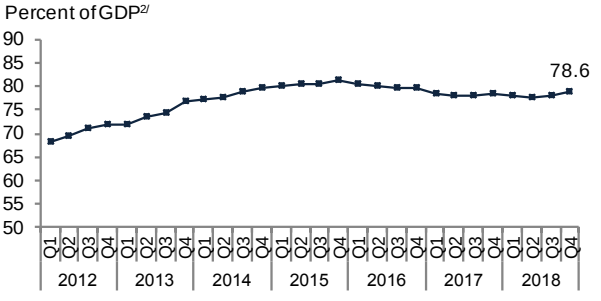


Note: ( ) represents percentage of unrated bonds to total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: household sector

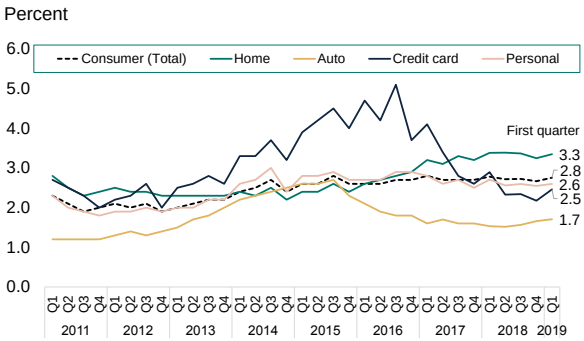
The ratio of household debt to GDP in the fourth quarter of 2018 rose from the previous quarter, and was expected to continue increasing. Overall credit quality of consumer loans, as indicated by the NPL ratio, remained stable. Nevertheless, debt serviceability for certain loan types continued to deteriorate, as reflected by the NPL new-entry rates of mortgage and auto loans that continued to trend up.

Household debt to GDP[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
[3/] Household debt and GDP data are revised. This results in the different debt to GDP ratios compared to the last MPR.
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

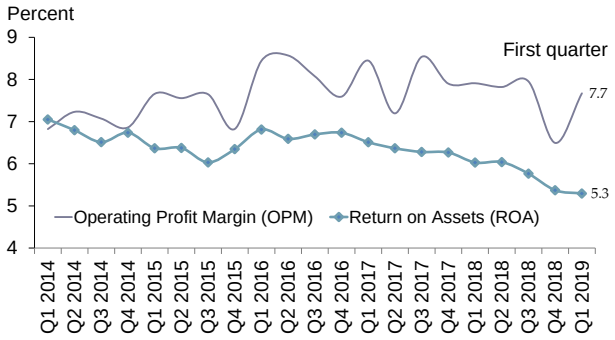


Source: Bank of Thailand

Stability: corporate sector

Large corporates maintained their high profitability. Meanwhile, debt serviceability of SMEs would need to be monitored, particularly small businesses in certain sectors with vulnerable financial positions. This was reflected by the operating profit margin and interest coverage ratio of some small businesses that remained negative. The NPL ratio among SMEs stood at 4.6 percent, above the historical average.

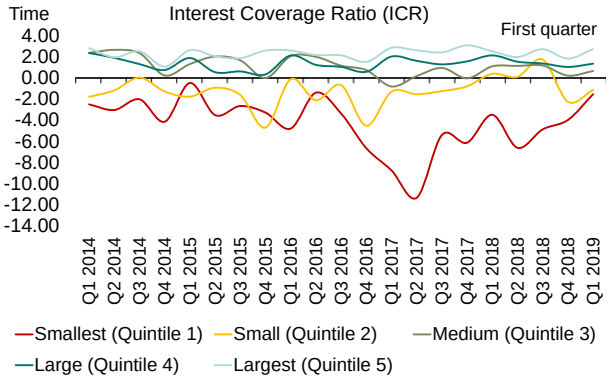
Operating Profit Margin (OPM) and Return on Assets (ROA)*



Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.

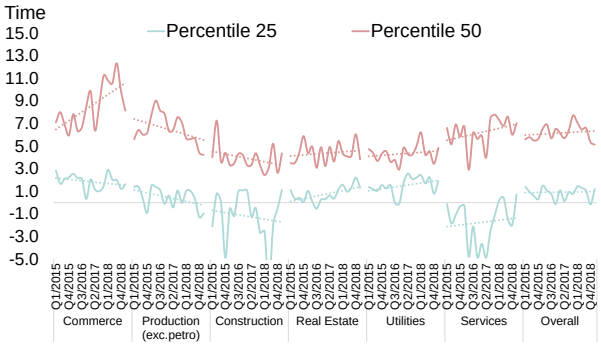
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Debt serviceability at 25th percentile of each group of firm size



Source: Stock Exchange of Thailand, calculation by Bank of Thailand

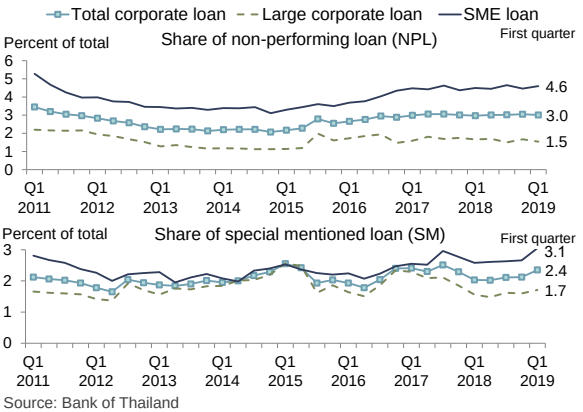
Interest Coverage Ratio, classified by sectors



Note: * production exclude Petroleum and chemicals

Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

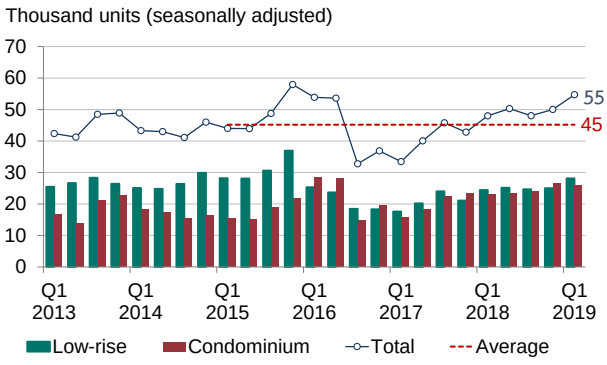


Source: Bank of Thailand

Stability: real estate

Foreign demand in the real estate market continued to decrease in response to the global economic slowdown, which could lengthen time to sell existing excess supply. In addition, the acceleration in residential ownership transfers, and a delay in new project launches by developers, could be observed prior to the implementation of the revised LTV ratio measures. Meanwhile, housing price index remained largely stable.

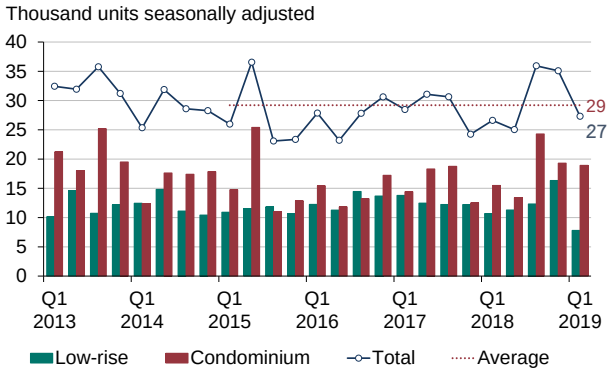
Residential transfer units in Bangkok and its vicinity



Note: *Average during 2014-2017

Source: Real Estate Information Center

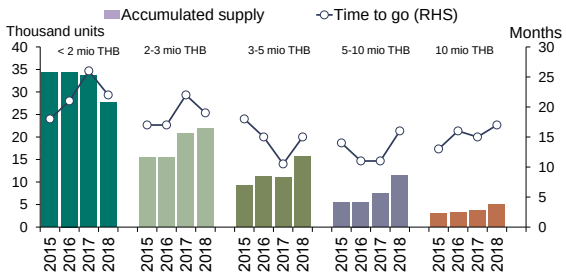
New residential projects launched in Bangkok and its vicinity



Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand

Note: *Average during 2014-2017

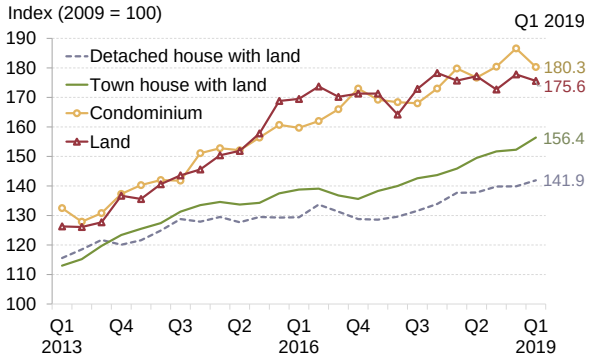
Excess supply of condominium in Bangkok and vicinity by price level and 'Time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold out at the average sales rate per month (since projects launched) given no additional supply.

Source: AREA and calculation by the Bank of Thailand

Housing price indices



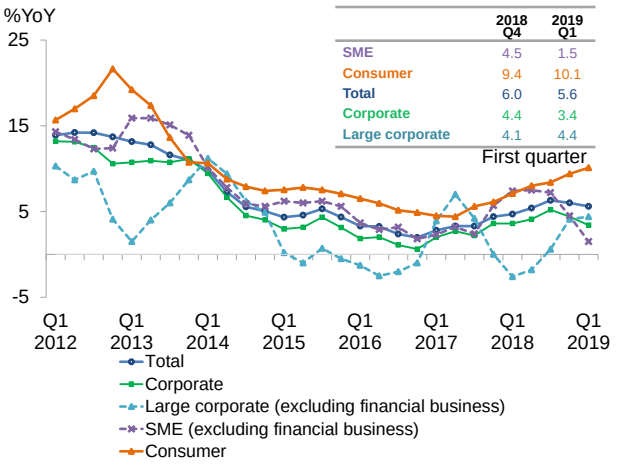
Note: Calculation based on commercial bank loan data

Source: Bank of Thailand

Stability: financial institutions

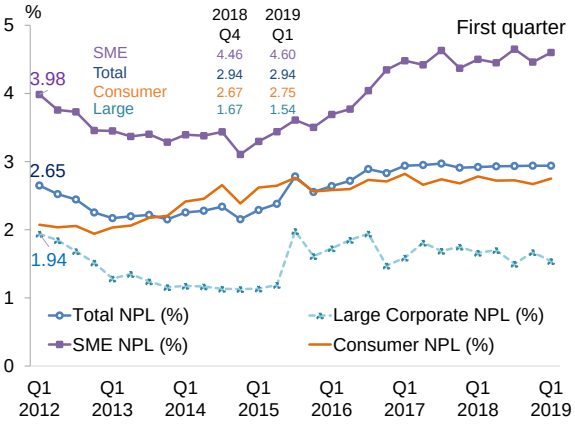
Financial institutions maintained strong financial positions, as reflected in high levels of capital buffers among commercial banks to cushion against risks should credit quality deteriorate. In the first quarter of 2019, commercial bank credits continued to expand mainly due to consumer loans. Meanwhile, overall NPL ratio stabilised at a high level, especially for loans extended to SMEs.

Credit growth in the commercial bank system



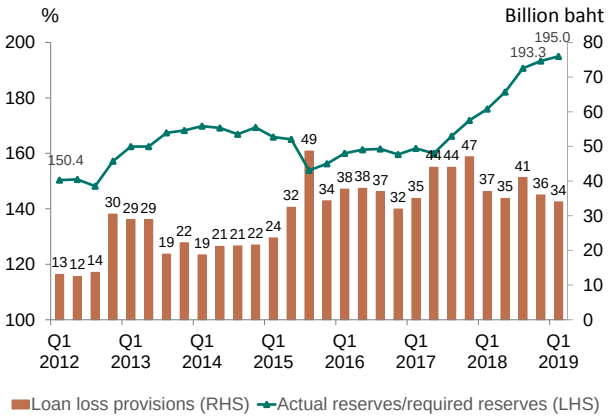
Source: Bank of Thailand

Non-performing loan (NPL)



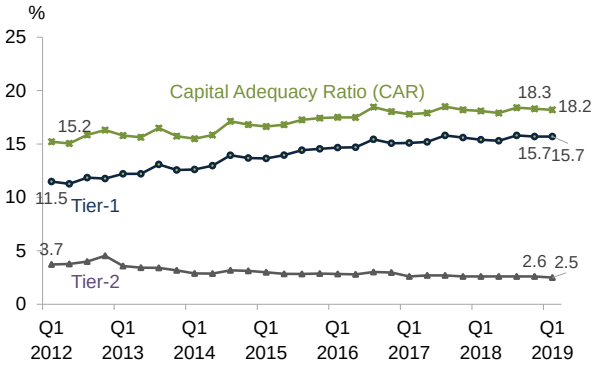
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

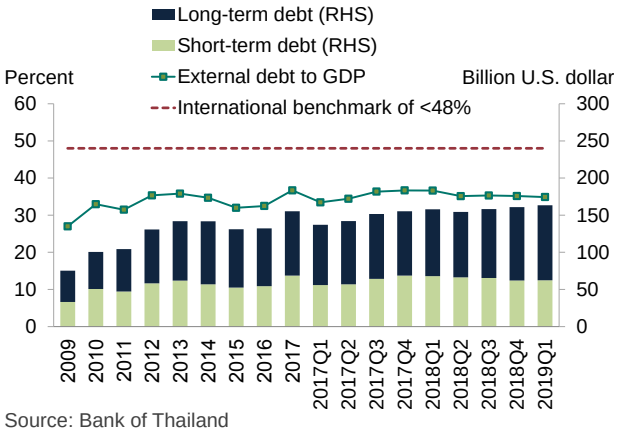


Source: Bank of Thailand

Stability: external positions

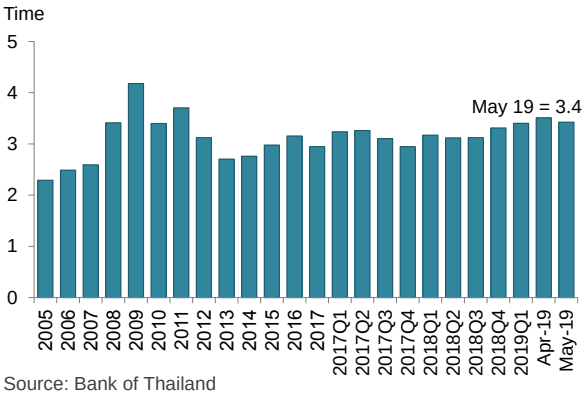
Thailand's external stability remained sound, as reflected in high levels of international reserves and a sustained current account surplus. Moreover, the ratio of external debt to GDP was below an international benchmark. This would help the Thai economy to be resilient against volatilities in global financial market.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

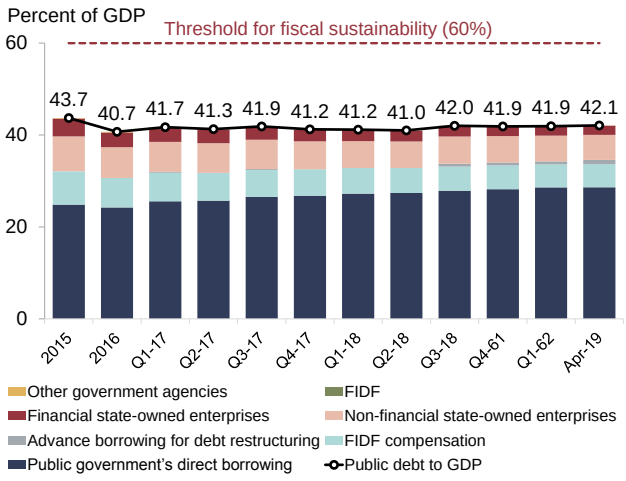


Source: Bank of Thailand

Stability: fiscal sector

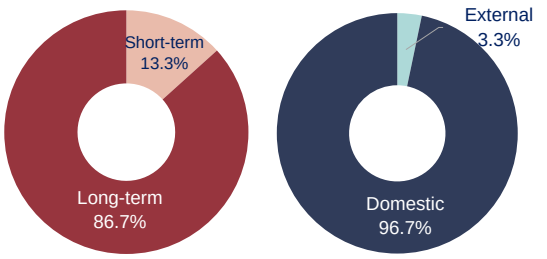
Fiscal stability remained sound. The ratio of public debt to GDP stayed below the sustainability threshold.

Public debt to GDP



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of April 2019



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office


Pursuing Sustainable
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